BOT-BASED SYSTEM FOR PERSONALIZED AND HOSPITRECOMMENDATIONS IN CARDIOVASCULAR DISEASE MANAGEMENT

PROJECT EVALUATION I REPORT

Submitted by

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degree

of

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in

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM

TIRUCHIRAPPALLI – 621 112.

DECEMBER 2023

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

BONAFIDE CERTIFICATE

Certified that this project report **"BOT-BASED SYSTEM FOR PERSONALIZED MEDICINE AND HOSPITAL RECOMMENDATIONS IN CARDIOVASCULAR DISEASE MANAGEMENT"** is the Bonafide work of **"BRIDJITH CLEMENTSIA.A, JANANI.D, NITHYA SRI.P, VARSHIKA.K "** who carried out the project work during the academic year 2023-2024 under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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DECLARATION

I hereby declare that the work entitled **''BOT-BASED SYSTEM FOR PERSONALIZED MEDICINE AND HOSPITAL RECOMMENDATIONS IN CARDIOVASCULAR DISEASE MANAGEMENT''** is submitted in partial fulfillment of the requirement for the reward of the degree in B.Tech**., K.Ramakrishnan College of Engineering (Autonomous),**Trichy, is a record of our own work carried out by me during the academic year 2023-2024 under the supervision and guidance of **Mrs.M.R. NITHYA, M.E.,** Assistant Professor, Department of Computer Science and Business Systems, K.Ramakrishnan College of Engineering (Autonomous). The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has not been submitted for the award of any degree or diploma, either in this or any other University.

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BOT-BASED SYSTEM FOR PERSONALIZED MEDICINE AND HOSPITAL RECOMMENDATIONS IN CARDIOVASCULAR DISEASE MANAGEMENT

ABSTRACT

The Heart is one of the most vital structures in the human body. It is the center of the circulatory system. Heart disease is a main life intimidating disease that can origin either death or a severe long-term disability. However, there is lack of effective tools to discover hidden relationships and trends in e-health data. Medical diagnosis is a complex task and plays a dynamic role in saving human lives so it needs to be executed accurately and efficiently. A suitable and precise computer based automated decision support system is required to reduce cost for achieving clinical tests. Health analytics have been proposed using ML to predict accurate patient data analysis. The data produced from health care industry is not mined. Data mining techniques can be used to build an intelligent model in medical field using data sets which involves risk factor of patients. The knowledge discovery in database (KDD) is startled with development of approaches and techniques for making use of data. This thesis provides an insight into deep learning and machine learning techniques used in diagnosing diseases.

Numerous data mining classifiers have been conversed which has emerged in recent years for efficient and effective disease diagnosis. This thesis proposes a heart attack prediction system using Deep learning techniques, specifically Multi-Layer Perceptron (MLP*)* to predict the likely possibilities of heart related diseases of the patient. MLP is a very powerful classification algorithm that makes use of Deep Learning approach in Artificial Neural Network. The proposed model incorporates deep learning and data mining to provide the accurate results with minimum errors. As the technology provides a clear and enhanced understanding about the prediction of the accurate and precise healthcare analysis and prediction.

Abstract keywords : Cardiovascular disease(CVD), Machine learning, Intelligent automated system, Predictive diagnosis, Multi-layer Perceptron Neural Network.

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LIST OF ABBREVIATIONS

SERIAL NO.	ABBREVIATION	FULL FORM
1	CVD	Cardiovascular Disease
2	SRS	Software Requirements Specification
3	ML	Machine Learning
4	NLP	Natural Language Processing
5	KNN	K-Nearest Neighbour Algorithm
6	GA	Genetic Algorithm
7	NB	Naive Bayes
8	SVM	Support Vector Machine
9	CHD	Coronary Heart Disease
10	DNN	Deeper Neural Network
11	PSO	Particle Swarm Optimization
12	MLP	Multi-layer Perceptron
13	ROC	Receiver Operating Characteristic

APPENDICES

CHAPTER 1

1. INTRODUCTION

1.1 OVERVIEW

Cardiovascular diseases (CVDs) continue to be a formidable global health challenge, particularly prevalent in industrialized nations, where their impact is felt through a significant toll on lives and quality of life. This project aims to address the pressing need for accurate and efficient prediction of cardiovascular diseases through the utilization of advanced technologies, specifically focusing on a deep learning algorithm known as the Multilayer Perceptron. The Software Requirements Specification (SRS) presented herein outlines a comprehensive framework designed not only to predict CVDs with improved accuracy but also to provide essential prescription details and recommend suitable healthcare professionals. In the face of preventable yet rising instances of CVDs, there exists a critical gap in preventive measures. The complexity of diagnosing heart diseases necessitates a sophisticated approach, and recent strides in machine learning provide a promising avenue.

This project not only leverages these advancements but also explores the implementation of intelligent automated systems to supportmedical practitioners in predictive diagnosis and decision-making. zThe focus on the MultilayerPerceptron Neural Network with Back-propagation as the training algorithm underscores the commitment to enhancing the accuracy of the diagnostic system. The iterative application of the back-propagation algorithm, as demonstrated in the study, aims to minimize error rates andthereby improve the overall diagnostic accuracy of the system. Beyond prediction, the projectenvisions a holistic approach by extending its capabilities to provide detailed prescription information related to the predicted diseases and recommending healthcare professionals. By addressing these facets, the project seeks to contribute to the global effort in combating cardiovascular diseases, fostering timely medical care, and ultimately saving more effective, efficient and lifesaving treatments for cardiovascular diseases and problems.

1.2 PROBLEM STATEMENT

Heart disease is a complex condition that can be caused by a variety of factors, including lifestyle choices, genetics, and underlying health conditions. As a result, accurately predicting heart disease risk requires a thorough analysis of a patient's medical history, physical examination, and diagnostic tests. Deep learning algorithms offer several advantages over traditional approaches to heart disease prediction. For example, these algorithms can analyse large datasets of patient information, including medical images and clinical data, to identify patterns and relationships that may not be apparent to human experts. Moreover, deep learning models can learn from large datasets without being explicitly programmed, allowing them to adapt and improve over time as new data becomes available. This makes deep learning algorithms a valuable tool for predicting heart disease risk and developing personalized treatment plans for patients. Several deep learning algorithms have been proposed for heart disease prediction, including deep neural networks, convolutional neural networks, and recurrent neural networks. These algorithms use various techniques to extract relevant features from medical data, including signal processing, image analysis, and natural language processing. One key challenge in developing deep learning models for heart disease prediction is ensuring that the models are accurate, reliable, and clinically relevant. This requires rigorous validation of the models using independent datasets and clinical trials to ensure that they are suitable for use in a clinical setting. Despite these challenges, deep learning algorithms offer significant potential for improving the accuracy and efficiency of heart disease prediction, enabling earlier diagnosis and more effective treatment for patients. As such, these algorithms are likely to play an increasingly important role in the future of cardiovascular medicine.

1.3 AIM

The aim of heart disease prediction using deep learning algorithms are as follows:

- Early detection: One of the primary objectives of heart disease prediction using deep learning algorithms is to detect the condition at an early stage. Early detection of heart disease can improve patient outcomes and prevent life-threatening complications.
- Accurate diagnosis: Deep learning algorithms can help healthcare professionals accurately diagnose heart disease by analyzing large datasets of patient information, including medical images and clinical data.

1.4 RESEARCH OBJECTIVES

- Personalized treatment: Deep learning algorithms can help healthcare professionals develop personalized treatment plans for patients based on their unique risk factors, medical history, and other clinical parameters.
- Improved efficiency: Deep learning algorithms can automate many aspects of heart disease prediction, enabling healthcare professionals to analyze large amounts of patient data more quickly and efficiently.
- Reduced healthcare costs: By improving the accuracy and efficiency of heart disease prediction, deep learning algorithms can help reduce healthcare costs by enabling earlier diagnosis and more effective treatment.
- Better patient outcomes: Ultimately, the goal of heart disease prediction using deep learning algorithms is to improve patient outcomes by enabling earlier detection, more accurate diagnosis, and more personalized treatment.

1.5 RESEARCH GAP IDENTITY

The utilization of Multilayer Perceptron (MLP) algorithms for heart disease prediction serves the primary purpose of constructing a sophisticated and accurate predictive model that aids in the early detection and assessment of cardiovascular conditions. MLP, being a type of artificial neural network, is well-suited for this task due to its proficiency in recognizing intricate patterns within complex datasets related to heart health. One crucial objective is the recognition of subtle correlations among various health indicators, symptoms, and patient profiles, enabling the identification of early warning signs associated with heart diseases. MLP's inherent ability to model non-linear relationships and its multilayer architecture contribute to capturing the intricate dynamics of heart health, where risk factors often exhibit complex interdependencies. Additionally, MLP algorithms possess the capacity to automatically learn and extract relevant features from diverse types of input data, facilitating the identification of key factors influencing heart disease. In essence, the application of MLP algorithms in heart disease prediction aims to enhance accuracy by effectively leveraging the network's capabilities in pattern recognition, feature extraction, and adaptability to complex data relationships. The scope of the heart disease prediction project using an MLP algorithm is extensive, with profound implications for healthcare. At its core, the project aims to make a substantial impact by offering a predictive tool capable of early detection of heart diseases.

CHAPTER 2

2. LITERATURE SURVEY

2.1 TITLE: APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN CARDIOVASCULAR IMAGING

AUTHORS: MAXIME SERMESANT AND HERVE DELINGETTE

YEAR: 2023

Image registration is an essential but challenging task in medical image computing, especially for echocardiography, where the anatomical structures are relatively noisy compared to other imaging modalities. Traditional (non-learning) registration approaches rely on the iterative optimization of a similarity metric which is usually costly in time complexity. In recent years, convolutional neural network (CNN) based image registration methods have shown good effectiveness. In the meantime, recent studies show that the attention-based model (e.g., Transformer) can bring superior performance in pattern recognition tasks.

We provide experiments on 2D-echocardiography registration to answer the former question partially and provide a benchmark solution. Our results on a large public 2D-echocardiography dataset show that the patch-based MLP/Transformer model can be effectively used for unsupervised echocardiography registration. They demonstrate comparable and even better registration performance than a popular CNN registration model. In particular, patch-based models better preserve volume changes in terms of Jacobian determinants, thus generating robust registration fields with less unrealistic deformation. Our results demonstrate that patch-based learning methods, whether with attention or not, can perform high-performance unsupervised registration tasks with adequate time and space complexity.

DISADVANTAGES

- Unclear echocardiography registration with MLPs, transformers; interpretability, data dependency.

2.2 TITLE: MACHINE LEARNING-BASED CORONARY ARTERY DISEASE DIAGNOSIS COMPREHENSIVE REVIEW

AUTHORS: ALIZADEHSANIROOHALLAH , MOLOUD ABDAR , MOHAMD ROSHANZ

YEAR: 2023

ML-based techniques have been successfully applied on various types of CAD datasets. These algorithms have demonstrated promising performance in the detection and treatment of CAD. This study comprehensively reviews how, when, and where ML techniques have been applied for CAD detection. By examining the works that have been published since 1992, this study aims at addressing the following: (a) various ML techniques that have been applied and their pros and cons, (b) the dataset characteristics (features and size) and how they impact the overall performance, and (c) performance achieved by each ML method for each dataset. The findings obtained in this study may assist researchers, physicians, governments and patients to take necessary steps to improve the quality of life. Both researchers and physicians are seeking economical, accurate and fast CAD diagnosis/treatment solutions.

This review paper will identify the best results and weaknesses of previous studies so that they can better deal with various challenges in their future works. In other words, this study identifies several strengths and weaknesses of previous data and methods. Additionally, current research benefits governments and patients. The accuracy, cost and speed of diagnosis/treatment of CAD are important issues for both of these groups. Thus, providing the best results and methods can be helpful for these individuals. Moreover, the ramifications of current research extend beyond the confines of the laboratory and medical institutions. Governments and patients, both integral stakeholders in the healthcare ecosystem, stand to benefit significantly from advancements in CAD research. The triad of accuracy, cost-effectiveness, and expeditiousness in the diagnosis and treatment of Coronary Artery Disease (CAD) emerges as a pivotal focus. The significance of this trinity is underscored by its direct impact on the efficiency of healthcare delivery, financial burden, and overall patient outcomes.

DISADVANTAGES

- Support Specific datasets only.

2.3 TITLEPSO ALGORITHM FOR RULE OPTIMIZATION IN HEART DISEASE DIAGNOSIS COMPARATIVE STUDY

AUTHORS: SMITH, A, JOHNSON, B. CHEN

YEAR: 2023

The task of classification becomes very difficult when the number of possible various combinations of parameters is so high. The self-adaptability of evolutionary algorithms depended on population is very useful in rule extraction and selection for data mining. In recent years, evolutionary algorithms; such as genetic algorithm, PSO algorithm and ant colony algorithm have appeared as necessary procedures to look for valuable and interesting information from databases. Particularly, there are many efforts to apply genetic algorithms (GAs) in data mining to classification tasks. Moreover, the particle swarm optimization (PSO) algorithm, which has arisen as a new meta-heuristic algorithm derived from nature, has involved many scholars' interests.

The use of swarm intelligence in data mining is not always easy due to its heavy computational load specifically in the case of a great data set and the difficulty of the domain. In this paper, we examine the use of PSO algorithm for production of rules in heart disease from original datasets and optimization of these rules and producing the best rules with PSO algorithm. The optimization target is based on the accuracy and the complexity of selected rules through experiments on some well-known benchmark data sets from the University of California, Irvine (UCI) machine learning repository. Our objective is to produce best rules for diagnosis of the heart disease. Navigating the application of swarm intelligence in data mining encounters challenges, particularly in managing computational demands, especially with extensive datasets and complex domains. Our aim is to derive optimal rules that enhance the diagnostic precision of heart disease.

DISADVANTAGES

- Difficult to construct the best rules.

2.4 TITLE: A DATA MINING MODEL FOR PREDICTING THE CORONARY HEART DISEASE USING RANDOM FOREST CLASSIFIER

AUTHORS: A. SHEIK ABDULLAH, A AUGUSTINE, NC AKHIL, OS DEEPA

YEAR: 2022

Data mining has become a fundamental methodology for computing applications in medical informatics. Progress in data mining applications and its implications are manifested in the areas of information management in healthcare organizations, health informatics, epidemiology, patient care and monitoring systems, assistive technology, large-scale image analysis to information extraction and automatic identification of unknown classes. Coronary Heart Disease refers to the failure of coronary circulation to supply adequate circulation to cardiac muscle and its surrounding tissue. This restricts the supply of blood and oxygen to the heart, particularly during exertion when the myocardial metabolic demands are increased. This paper focuses on the creation of a data mining model using the Random Forest classification algorithm for evaluating and predicting various events related to CHD.

Some of these studies, has made with the implementation of data mining algorithm such as K-NN, Naïve bayes, K-means, ID3, and Apriori algorithms. The growing healthcare burden and suffering due to life threatening diseases such as heart disease and the escalating cost of drug development can be significantly reduced by design and development of novel methods in data mining technologies and allied medical informatics disciplines. In CHD, if the risk factors are predicted in advance two sorts of problem can be solved. First, various surgical treatments such as angioplasty, coronary stents, coronary artery bypass and heart transplant can be avoided to a great extent. Second, the associated cost with each risk factor can be reduced. Using random forest algorithm, we obtained accuracy of 86.9% for prediction of heart disease with sensitivity value 90.6% and specificity value 82.7%. From the receiver operating characteristics, we obtained the diagnosis rate for prediction of heart disease using random forest is 93.3%.

DISADVANTAGES

- Tree structure become complex.

2.5 TITLE: EFFECTIVE HEART DISEASE PREDICTION USING HYBRID MACHINE LEARNING TECHNIQUES

AUTHORS: CHANDRASEGAR THIRUMALAI, GAUTAM SRIVASTAVA

YEAR: 2022

It is difficult to identify heart disease because of several contributory risk factors such as diabetes, high blood pressure, high cholesterol, abnormal pulse rate and many other factors. Various techniques in data mining and neural networks have been employed to find out the severity of heart disease among humans. The severity of the disease is classified based on various methods like K-Nearest Neighbor Algorithm (KNN), Decision Trees (DT), Genetic algorithm (GA), and Naive Bayes (NB). The nature of heart disease is complex and hence, the disease must be handled carefully.The far-reaching implications of contemporary research in the field of Computer-Aided Diagnosis (CAD) extend well beyond the sterile confines of laboratories and medical institutions.

This burgeoning area of study is poised to bring about transformative changes that resonate deeply with the core stakeholders in the intricate tapestry of the healthcare ecosystem—governments and patients. Governments, as stewards of public health, are keenly interested in the outcomes of CAD research. The potential benefits span a spectrum of critical areas, ranging from improved public health outcomes to the optimization of resource allocation. By embracing advancements in CAD, governments can bolster the efficiency of healthcare delivery systems, ensuring that diagnostic and treatment processes are not only accurate but also streamlined. This, in turn, has the potential to alleviate the strain on healthcare infrastructure, enhancing its overall resilience and responsiveness to the needs of the population.

DISADVANTAGES

- Only analyzed labeled datasets.

2.6 TITLE: AN INTEGRATED MACHINE LEARNING TECHNIQUES FOR ACCURATE HEART DISEASE PREDICTION

AUTHORS: AHMED AL AHDAL, MANIK RAKHRA, SUMIT BADOTRA, TAHA FADHAEEL

YEAR: 2022

Medical organizations, all around the world, collect data on various health related issues. These data can be exploited using various machine learning techniques to gain useful insights. But the data collected is very massive and, many a times, this data can be very noisy. A particularly noteworthy machine learning technique that has proven instrumental in predicting the presence or absence of heart-related diseases is the Support Vector Machine (SVM). As a widely acclaimed supervised machine learning approach, SVM operates with a pre-defined target variable, making it adept at both classification and prediction tasks. For classification purposes, SVM endeavours to identify a hyperplane within the feature space, strategically positioned to demarcate distinct classes. This hyperplane, essentially a multidimensional plane, seeks to maximize the margin between classes, optimizing the separation of data points.

The SVM model ingeniously represents training data points as coordinates in the feature space, leveraging a mapping technique that ensures points belonging to disparate classes are distinctly segregated by a maximized margin. This margin, the spatial gap between classes, serves as a critical criterion for classification accuracy. When faced with new or test data points, SVM employs the same mapping strategy to position them within the feature space, subsequently categorizing them based on their relative placement with respect to the margin. The efficacy of SVM in heart disease prediction lies in its ability to discern intricate patterns within extensive and noisy datasets. By strategically defining the boundaries between classes, SVM enhances the precision and reliability of predictions, offering a robust framework for advancing diagnostic capabilities in the realm of cardiovascular health.

DISADVANTAGES

- Computational complexity is high.

2.7 TITLE: DEVELOPMENT OF BIG DATA PREDICTIVE ANALYTICS MODEL FOR DISEASE PREDICTION USING MACHINE LEARNING TECHNIQUE

AUTHORS: R. VENKATESH, C. BALASUBRAMANIAN, M. KALIAPPAN

YEAR: 2021

Now days, health prediction in modern life becomes very much essential. Big data analysis plays a crucial role to predict future status of health and offers preeminent health outcome to people. Heart disease is a prevalent disease cause’s death around the world. A lot of research is going on predictive analytics using machine learning techniques to reveal better decision making. Big data analysis fosters great opportunities to predict future health status from health parameters and provide best outcomes. This work uses Naive Bayes classification technique to build an accuracy predictive model as it is one of the most commonly applied machines learning technique. The Naive Bayes methodology is recognized for its efficacy in handling extensive datasets, particularly within the realm of big data. In this study, we applied the Naive Bayes approach to train on heart disease data sourced from the UCI Machine Learning Repository. Subsequently, the model was deployed to make predictions on test data, aiming to ascertain disease classifications.

Our devised approach, termed BPA-NB, yielded commendable results, exhibiting a superior accuracy of 97.12% in predicting disease rates. Notably, the BPA-NB scheme was implemented utilizing Hadoop-Spark as a robust big data computing tool, thereby enabling the extraction of profound insights from healthcare data. The experiments conducted in this study aimed at forecasting diverse patients' future health conditions. Leveraging the training dataset, we estimated crucial health parameters essential for classification. The outcomes of our research underscore the potential of early disease detection, offering valuable insights into the anticipation of patients' future health trajectories

DISADVANTAGES

- Not in real time environments.

2.8 TITLE: DESIGNING AN EFFICIENT CONVOLUTIONAL NEURAL NETWORK FOR PREDICTING CORONARY HEART DISEASE

AUTHORS: TAMAL BATABYAL, MEHELI BASU, SCOTT T. ACTON

YEAR: 2020

In this research endeavour, we focus on the development of an advanced convolutional neural network (CNN) specifically tailored for the efficient prediction of coronary heart disease. Drawing on the formidable capabilities of deep learning, our proposed CNN architecture is intricately designed to strike an optimal balance between computational efficiency and predictive accuracy. The overarching goal of this study is to contribute a resilient and streamlined neural network model, adept at discerning subtle patterns within medical imaging data, thereby elevating the precision and timeliness of coronary heart disease prediction. This research marks a significant step forward in the amalgamation of cutting edge technology with healthcare, promising improved prognostication in the realm of cardiovascular health.

At the heart of our research lies the imperative to enhance predictive models for coronary heart disease, a pervasive and critical cardiovascular condition. Traditional diagnostic approaches often fall short in capturing the nuanced patterns present in medical imaging data, necessitating the application of sophisticated deep learning techniques. Convolutional Neural Networks, known for their prowess in image analysis, emerge as a promising avenue for addressing this challenge. Our CNN architecture is purposefully tailored to meet the unique demands of predicting coronary heart disease with a focus on efficiency and accuracy. The core strength of our proposed CNN architecture lies in its meticulous design, a result of a nuanced exploration of hyperparameters, layer configurations, and feature extraction strategies

DISADVANTAGES

- Convolutional Neural Network (CNN) may pose challenges in resource-intensive environments.

2.9 TITLE ARTIFICIAL LAMPYRIDAE CLASSIFIER (ALAC) FOR CORONARY ARTERY HEART DISEASE PREDICTION IN DIABETES PATIENTS

AUTHORS: B. NARASIMHAN, A. MALATHI

YEAR: 2019

Typically the essential contemplations of traditional computing are accuracy, sureness, and meticulousness. We recognize this as hard computing. Interestingly, the important thought in soft computing is that exactness and sureness convey a cost; sand that calculation, reasoning, and basic leadership should misuse (wherever conceivable) the resistance for imprecision, uncertainty, approximate reasoning, and incomplete truth for getting minimal effort arrangements. This prompts the wonderful human capacity of understanding mutilated discourse, translating messy penmanship, appreciating the subtleties of characteristic language, condensing content, perceiving and arranging pictures, driving a vehicle in thick rush hour gridlock, and, all the more for the most part, settling on normal choices in a domain of uncertainty and imprecision.

The test, at that point, is to misuse the resilience for imprecision by concocting strategies for calculation that lead to an acknowledge capable arrangement requiring little to no effort. This, generally, is the core value of soft computing. Heart disease remains the number one cause of death throughout the world for the past decades. In 2015, the World Health Organization (WHO) has estimated that 17.7 million deaths have occurred worldwide due to heart diseases. Heart diseases are the primary cause of death globally: more people die annually from heart disease than any other causes. If we can predict and provide warning beforehand, a handful of deaths can be prevented. The application of soft computing brings a new dimension to risk prediction.

DISADVANTAGES

- Less accuracy.

2.10 TITLE CORONARY HEART DISEASE INTERPRETATION BASED ON DEEP NEURAL NETWORK

AUTHORS: : ANNISA DARMAWAHYUNI, SITI NURMAINI, FIDARIUS

YEAR: 2019

Coronary heart disease (CHD) population increases every year with a significant number of deaths. Moreover, the mortality from coronary heart disease gets the highest prevalence in Indonesia at 1.5 percent. The misdiagnosis of coronary heart disease is a crucial fundamental that is the major factor that caused death. To prevent misdiagnosis of CHD, an intelligent system has been designed. This paper proposed a simulation which can be used to diagnose the coronary heart disease in better performance than the traditional diagnostic methods. Some researchers have developed a system using conventional neural network or other machine learning algorithm, but the results are not a good performance. Based on a conventional neural network, deeper neural network (DNN) is proposed to our model in this work. As known as, the neural network is a supervised learning algorithm that good in the classification task. Due to the number of CHD population increases every year with a significant number of deaths, we proposed a learning algorithm to get better performance in accuracy, sensitivity, and specificity in CHD interpretation.

CHD interpretation minimizes the diagnosis time of experts and increases the accuracy of diagnosis. As known as, the traditional feature-based classification is less effective since their performance is usually dependent on the quality of handcrafted features. Conquering those drawbacks, the exploration of feature learning is also used to improve the performance of traditional feature-based approaches. The significance of this approach lies in its potential to mitigate the shortcomings associated with traditional methods. By tapping into feature learning, our model endeavors to overcome the limitations posed by manually crafted features, providing a more adaptive and data-driven solution.

DISADVANTAGES

- Provide false positive rate.

CHAPTER 3

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Present days one of the major application areas of machine learning algorithms is medical diagnosis of diseases and treatment. Machine learning algorithms also used to find correlations and associations between different diseases. Nowadays many people are dying because of sudden heart attack. Prediction and diagnosing of heart disease becomes a challenging factor faced by doctors and hospitals both in India and abroad. In order to reduce number of deaths because of heart diseases, we have to predict whether person is at the risk of heart disease or not in advance. Data mining techniques and machine learning algorithms play a very important role in this area. Many researchers are carrying out their research in this area to develop software that can help doctors to take decision regarding both prediction and diagnosing of heart disease. In this paper we focused on how data mining techniques can be used to predict heart disease in advance such that patient is well treated. An important task of any diagnostic system is the process of attempting to determine and/or identify a possible disease or disorder and the decision reached by this process. For this purpose, machine learning algorithms are widely employed. For these machine learning techniques to be useful in medical diagnostic problems, they must be characterized by high performance, the ability to deal with missing data and with noisy data, the transparency of diagnostic knowledge, and the ability to explain decisions. As people are generating more data everyday so there is a need for such a classifier which can classify those newly generated data accurately and efficiently. This System mainly focuses on the supervised learning technique called the Random forests for classification of data by changing the values of different hyper parameters in Random Forests Classifier

3.1.1 DISADVANTAGES

- Labelled data-based disease classification
- Provide high number of false positive
- Binary classification can be occurred
- Computational complexity

3.2 PROPOSED SYSTEM

Cardiovascular disease continues to claim an alarming number of lives across the globe. CVD disease is the greatest scourge affecting the industrialized nations. CVD not only strikes down a significant fraction of the population without warning but also causes prolonged suffering and disability in an even larger number. Although large proportion of CVDs is preventable, they continue to rise mainly because preventive measures are inadequate. Heart disease diagnosis has become a difficult task in the field of medicine. This diagnosis depends on a thorough and accurate study of the patient's clinical tests data on the health history of an individual. The tremendous improvement in the field of machine learning aim at developing intelligent automated systems which helps the medical practitioners in predicting as well as making decisions about the disease. Such an automated system for medical diagnosis would enhance timely medical care followed by proper subsequent treatment thereby resulting in significant lifesaving. Incorporating the techniques of classification in these intelligent systems achieve at accurate diagnosis. Neural Networks has emerged as an important method of classification. Multi-layer Perceptron Neural Network with Back-propagation has been employed as the training algorithm in this work. This project proposes a diagnostic system for predicting heart disease with improved accuracy. The propagation algorithm has been repeated until minimum error rate was observed. And it is quite evident from the results presented in the previous section that the accuracy rate is maximized.

3.2.1 ADVANTAGES

- Accuracy is high
- Parallel processing
- Multiple heart diseases are predicted
- Reduce number of false positive rate

CHAPTER 4

4. ARCHITECTURE DIAGRAM

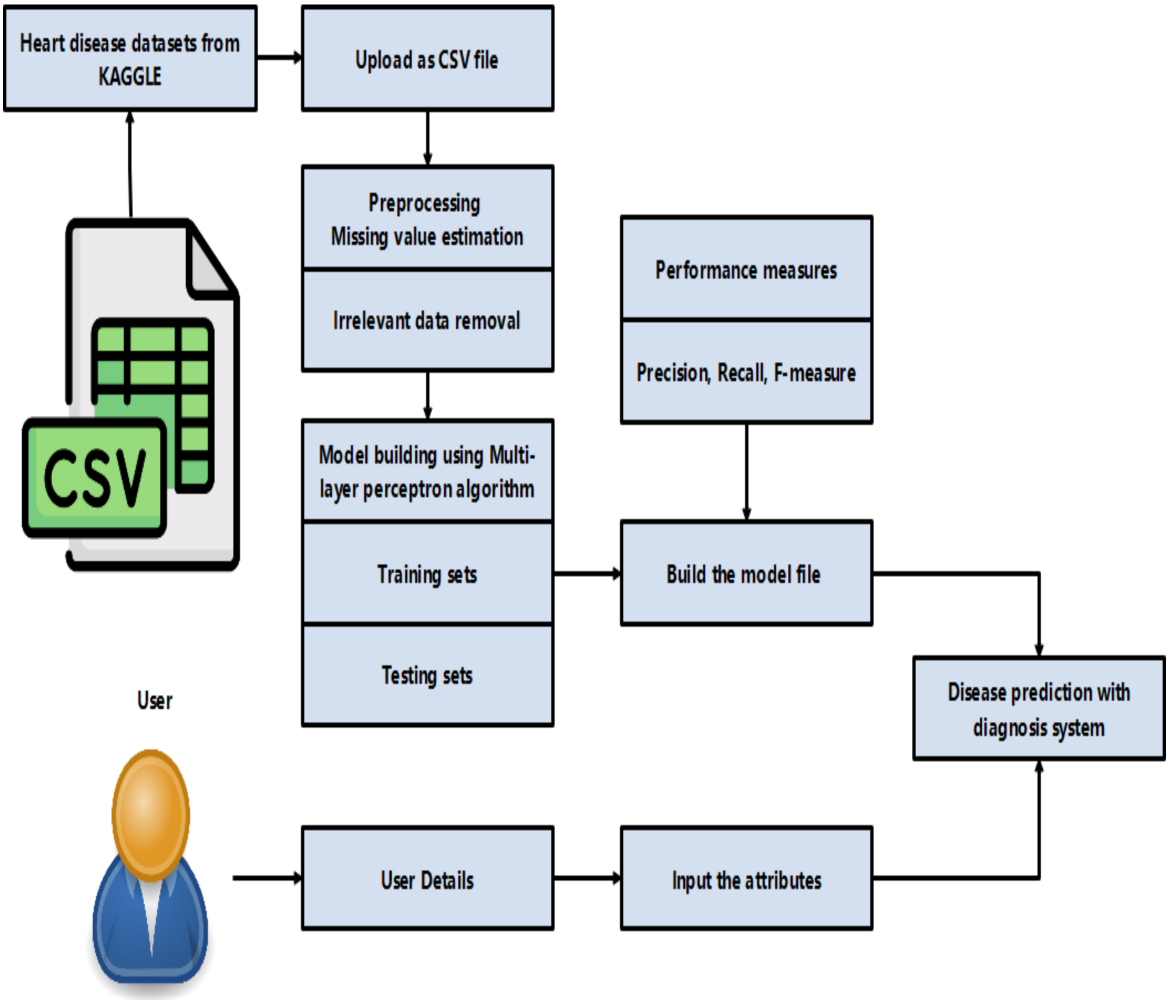


Fig 4.1 Architecture diagram

CHAPTER 5

5. SYSTEM REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

- Processor : Intel core processor 2.6.0 GHZ
- RAM : 1GB
- Hard disk : 160 GB
- Compact Disk : 650 Mb
- Keyboard : Standard keyboard
- Monitor : 15-inch colour monitor

5.2 SOFTWARE REQUIREMENTS

- Server Side : Python 3.7.4(64-bit) or (32-bit)
- Client Side : HTML, CSS, Bootstrap
- IDE : Flask 1.1.1
- Back end : MySQL 5.
- Server : WampServer 2i
- OS : Windows 10 64 –bit

5.3 TECHNOLOGY USED

5.3.1 PYTHON

Python is a high-level, interpreted programming language that is widely used in various domains such as web development, scientific computing, data analysis, artificial intelligence, machine learning, and more. It was first released in 1991 by Guido van Rossum and has since become one of the most popular programming languages due to its simplicity, readability, and versatility. One of the key features of Python is its easy-to-learn syntax, which makes it accessible to both novice and experienced programmers. It has a large standard library that provides a wide range of modules for tasks such as file I/O, networking, regular expressions, and more. Python also has a large and active community of developers who contribute to open-source libraries and packages that extend its capabilities. Python is an interpreted language, which means that it is executed line-by-line by an interpreter rather than compiled into machine code like C or C++. This allows for rapid development and testing, as well as easier debugging and maintenance of code. Python is used for a variety of applications, including web development frameworks such as Django and Flask, scientific computing libraries such as NumPy and Pandas, and machine learning libraries such as TensorFlow and PyTorch. It is also commonly used for scripting and automation tasks due to its ease of use and readability. Overall, Python is a powerful and versatile programming language that is widely used in a variety of domains due to its simplicity, ease of use, and active community.

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace. It provides constructs that enable clear programming on both small and large scales. In July 2018, Van Rossum stepped down as the leader in the language community. Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library. Python interpreters are available for many operating systems. CPython, the reference implementation of Python, is open-source software and has a community-based development model, as do nearly all of Python's other implementations. Python and CPython are managed by the non-profit Python Software Foundation. Rather than having all of its functionality built into its core, Python was designed to be highly extensible. This compact modularity has made it particularly popular as a means of adding programmable interfaces to existing applications. Van Rossum's vision of a small core language with a large standard library and easily extensible interpreter stemmed from his frustrations with ABC, which espoused the opposite approach. While offering choice in coding methodology, the Python philosophy rejects exuberant syntax (such as that of Perl) in favor of a simpler, less-cluttered grammar. As Alex Martelli put it: "To describe something as 'clever' is not considered a compliment in the Python culture."Python's philosophy rejects the Perl "there is more than one way to do it" approach to language design in favour of "there should be one—and preferably only one—obvious way to do it".

Python's developers strive to avoid premature optimization, and reject patches to non-critical parts of CPython that would offer marginal increases in speed at the cost of clarity. [When speed is important, a Python programmer can move time-critical functions to extension modules written in languages such as C, or use PyPy, a just-in-time compiler. CPython is also available, which translates a Python script into C and makes direct C-level API calls into the Python interpreter. An important goal of Python's developers is keeping it fun to use. This is reflected in the language's name a tribute to the British comedy group Monty Python and in occasionally playful approaches to tutorials and reference materials, such as examples that refer to spam and eggs (from a famous Monty Python sketch) instead of the standard for and bar. A common neologism in the Python community is pythonic, which can have a wide range of meanings related to program style. To say that code is pythonic is to say that it uses Python idioms well, that it is natural or shows fluency in the language, that it conforms with Python's minimalist philosophy and emphasis on readability. In contrast, code that is difficult to understand or reads like a rough transcription from another programming language is called unpythonic. Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed. Often, programmers fall in love with Python because of the increased productivity it provides. Since there is no compilation step, the edit-test-debug cycle is incredibly fast. Debugging Python programs is easy: a bug or bad input will never cause a segmentation fault. Instead, when the interpreter discovers an error, it raises an exception. When the program doesn't catch the exception, the interpreter prints a stack trace.

Python also has a large and active community of developers who contribute to a wide range of open-source libraries and tools, making it easy to find and use pre-built code to solve complex problems.

Python has a wide range of applications, including:

Data Science: Python is one of the most popular languages for data science, thanks to libraries like NumPy, Pandas, and Matplotlib that make it easy to manipulate and visualize data.

Machine Learning: Python is also widely used in machine learning and artificial intelligence, with libraries like TensorFlow, Keras, and Scikit-learn that provide powerful tools for building and training machine learning models.

Web Development: Python is commonly used in web development, with frameworks like Django and Flask that make it easy to build web applications and APIs.

Scientific Computing: Python is used extensively in scientific computing, with libraries like SciPy and SymPy that provide powerful tools for numerical analysis and symbolic mathematics.

In addition to its versatility and ease of use, Python is also known for its portability and compatibility. Python code can be run on a wide range of platforms, including Windows, macOS, and Linux, and it can be integrated with other languages like C and Java.

Overall, Python is a powerful and versatile programming language that is well-suited for a wide range of applications, from data science and machine learning to web development and scientific computing. Its simplicity, readability, and large community of developers make it an ideal choice for beginners and experts alike.

There are two attributes that make development time in Python faster than in other programming languages:

1. Python is an interpreted language, which precludes the need to compile code before executing a program because Python does the compilation in the background. Because Python is a high-level programming language, it abstracts many sophisticated details from the programming code. Python focuses so much on this abstraction that its code can be understood by most novice programmers.

2. Python code tends to be shorter than comparable codes. Although Python offers fast development times, it lags slightly in terms of execution time. Compared to fully compiling languages like C and C++, Python programs execute slower. Of course, with the processing speeds of computers these days, the speed differences are usually only observed in benchmarking tests, not in real-world operations. In most cases, Python is already included in Linux distributions and Mac OS X machines.

One of the strengths of Python is its rich ecosystem of third-party libraries and tools. These libraries provide a wide range of functionality, from scientific computing and data analysis to web development and machine learning. Some popular Python libraries and frameworks include:

NumPy: a library for numerical computing in Python, providing support for large, multi-dimensional arrays and matrices, along with a large collection of mathematical functions to operate on these arrays.

Pandas: a library for data manipulation and analysis in Python, providing support for reading and writing data in a variety of formats, as well as powerful tools for manipulating and analyzing data.

Matplotlib: a plotting library for Python that provides a variety of visualization tools, including line plots, scatter plots, bar plots, and more.

TensorFlow: an open-source machine learning library for Python that provides a variety of tools and algorithms for building and training machine learning models.

Django: a popular web framework for Python that provides a full-stack framework for building web applications, with support for everything from URL routing to user authentication and database integration.

Python's popularity has also led to a large and active community of developers who contribute to open-source projects and share code and resources online. This community provides a wealth of resources for learning Python, including tutorials, online courses, and forums for asking and answering questions.

Overall, Python is a versatile and powerful programming language that is well-suited for a wide range of applications. Its simplicity, flexibility, and wide range of libra

5.3.2 TENSORFLOW LIBARIES IN PYTHON

TensorFlow is an open-source machine learning framework developed by Google Brain Team. It is one of the most popular libraries for building and training machine learning models, especially deep neural networks. TensorFlow allows developers to build complex models with ease, including image and speech recognition, natural language processing, and more. One of the key features of TensorFlow is its ability to handle large-scale datasets and complex computations, making it suitable for training deep neural networks. It allows for parallelization of computations across multiple CPUs or GPUs, allowing for faster training times. TensorFlow also provides a high-level API called Keras that simplifies the process of building and training models. TensorFlow offers a wide range of tools and libraries that make it easy to integrate with other Python libraries and frameworks. It has built-in support for data preprocessing and visualization, making it easy to prepare data for training and analyze model performance. One of the major advantages of TensorFlow is its ability to deploy models to a variety of platforms, including mobile devices and the web.

Graph-based computation: TensorFlow uses a graph-based computation model, which allows for efficient execution of computations across multiple devices and CPUs/GPUs.

Automatic differentiation: TensorFlow provides automatic differentiation, which allows for efficient computation of gradients for use in backpropagation algorithms.

High-level APIs: TensorFlow provides high-level APIs, such as Keras, that allow developers to quickly build and train complex models with minimal code.

Preprocessing and data augmentation: TensorFlow provides a range of tools for preprocessing and data augmentation, including image and text preprocessing, data normalization, and more.

Distributed training: TensorFlow supports distributed training across multiple devices, CPUs, and GPUs, allowing for faster training times and more efficient use of resources.

Model deployment: TensorFlow allows for easy deployment of models to a variety of platforms, including mobile devices and the web.

Visualization tools: TensorFlow provides a range of visualization tools for analyzing model performance, including TensorBoard, which allows for real-time visualization of model training and performance.

5.3.3 PYCHARM

PyCharm is an integrated development environment (IDE) for Python programming language, developed by JetBrains. PyCharm provides features such as code completion, debugging, code analysis, refactoring, version control integration, and more to help developers write, test, and debug their Python code efficiently. PyCharm is available in two editions: Community Edition (CE) and Professional Edition (PE). The Community Edition is a free, open-source version of the IDE that provides basic functionality for Python development. The Professional Edition is a paid version of the IDE that provides advanced features such as remote development, web development, scientific tools, database tools, and more. PyCharm is available for Windows, macOS, and Linux operating systems. It supports Python versions 2.7, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, and 3.10.

Features:

- Intelligent code completion
- Syntax highlighting
- Code inspection
- Code navigation and search
- Debugging
- Testing
- Version control integration
- Web development support
- Scientific tools support
- Database tools support

Integration with other JetBrains tools

PyCharm's code completion feature can help speed up development by automatically suggesting code based on context and previously written code. It also includes a debugger that allows developers to step through code, set breakpoints, and inspect variables. PyCharm has integration with version control systems like Git, Mercurial, and Subversion. It also supports virtual environments, which allow developers to manage different Python installations and packages in isolated environments. The IDE also has features specifically geared towards web development, such as support for popular web frameworks like Django, Flask, and Pyramid. It includes tools for debugging, testing, and profiling web applications. PyCharm also provides scientific tools for data analysis, visualization, and scientific computing, such as support for NumPy, SciPy, and matplotlib. It also includes tools for working with databases, such as PostgreSQL, MySQL, and Oracle. Overall, PyCharm is a powerful and feature-rich IDE that can greatly increase productivity for Python developers.

Customization:

PyCharm allows developers to customize the IDE to their liking. Users can change the color scheme, fonts, and other settings to make the IDE more comfortable to use. PyCharm also supports plugins, which allow developers to extend the IDE with additional features.

Collaboration:

PyCharm makes it easy for developers to collaborate on projects. It supports integration with popular collaboration tools such as GitHub, Bitbucket, and GitLab. It also includes features for code reviews, task management, and team communication.

Education:

PyCharm provides a learning environment for Python programming language. PyCharm Edu is a free, open-source edition of PyCharm that includes interactive courses and tutorials for learning Python. It provides an easy-to-use interface for beginners and includes features such as code highlighting, autocompletion, and error highlighting.

Support:

PyCharm has an active community of users who provide support through forums and social media. JetBrains also provides comprehensive documentation, tutorials, and training courses for PyCharm. For users who need more personalized support, JetBrains offers a paid support plan that includes email and phone support.

Pricing:

PyCharm Community Edition is free and open-source. PyCharm Professional Edition requires a paid license, but offers a 30-day free trial. JetBrains also offers a subscription-based pricing model that includes access to all JetBrains IDEs and tools.

Integrations:

PyCharm integrates with a wide range of tools and technologies commonly used in Python development. It supports popular Python web frameworks like Flask, Django, Pyramid, and web2py. It also integrates with tools for scientific computing like NumPy, SciPy, and pandas. PyCharm also supports popular front-end technologies such as HTML, CSS, and JavaScript.

Performance:

PyCharm is known for its fast and reliable performance. It uses a combination of static analysis, incremental compilation, and intelligent caching to provide fast code completion and navigation. PyCharm also has a memory profiler that helps identify and optimize memory usage in Python applications.

Ease of Use:

PyCharm provides an intuitive and easy-to-use interface for developers. It has a well-organized menu structure, clear icons, and easy-to-navigate tabs. PyCharm also provides a variety of keyboard shortcuts and customizable keymaps that allow users to work efficiently without constantly switching between the mouse and keyboard.

Community:

PyCharm has a large and active community of developers who contribute to the development of the IDE. The PyCharm Community Edition is open-source, which means that anyone can contribute to its development. The PyCharm user community is also active in providing support, tips, and tutorials through forums, blogs, and social media.

5.4 MY SQL

MySQL is the world's most used open source relational database management system (RDBMS) as of 2008 that run as a server providing multi-user access to a number of databases. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation.

MySQL is a popular choice of database for use in web applications, and is a central component of the widely used LAMP open source web application software stack—LAMP is an acronym for "Linux, Apache, MySQL, Perl/PHP/Python." Free-software-open source projects that require a full-featured database management system often use MySQL.For commercial use, several paid editions are available, and offer additional functionality. Applications which use MySQL databases include: TYPO3, Joomla, Word Press, phpBB, MyBB, Drupal and other software built on the LAMP software stack. MySQL is also used in many high-profile, large-scale World Wide Web products, including Wikipedia, Google(though not for searches), Imagebook Twitter, Flickr, Nokia.com, and YouTube.

Inter images

MySQL is primarily an RDBMS and ships with no GUI tools to administer MySQL databases or manage data contained within the databases. Users may use the included command line tools, or use MySQL "front-ends", desktop software and web applications that create and manage MySQL databases, build database structures, back up data, inspect status, and work with data records. The official set of MySQL front-end tools, MySQL Workbench is actively developed by Oracle, and is freely available for use.

Graphical

The official MySQL Workbench is a free integrated environment developed by MySQL AB that enables users to graphically administer MySQL databases and visually design database structures. MySQL Workbench replaces the previous package of software, MySQL GUI Tools. Similar to other third-party packages, but still considered the authoritative MySQL frontend, MySQL Workbench lets users manage database design & modeling, SQL development (replacing MySQL Query Browser) and Database administration (replacing MySQL Administrator).MySQL Workbench is available in two editions, the regular free and open source Community Edition which may be downloaded from the MySQL website, and the proprietary Standard Edition which extends and improves the feature set of the Community Edition.

CHAPTER 6

6. RESEARCH NOVELTY

In today's world, cardiovascular disease is one of the most prevalent illnesses. Over 17.7 million people worldwide pass away from heart disease each year, according to a report. Of these deaths, 7.4 million were attributed to coronary heart disease and 6.7 million to stroke, according to estimates. One of the worst illnesses that can hit at any time and without warning is heart disease, and most doctors are unable to foresee silent heart attacks. The lack of specialists and the rise in patients with misdiagnosed conditions necessitated the creation of an accurate cardiovascular disease prediction system. As a result, fresh methods for mining medical data and machine learning are being investigated and created. The main objective of this study is to identify the most pertinent characteristics for diagnosing silent heart attacks by using classification algorithms to identify important patterns and features in medical data. The use of an artificial neural network will result in even more precise outcomes. Although the creation of such a system is not new, existing systems have shortcomings and are not intended to detect the potential for silent heart attacks. This study's objective is to address these problems and recommend the addition of distinctive features in order to build a system that is more complete. The accuracy of the results produced by the present heart attack prediction systems is insufficient. The accuracy is being pushed to a certain degree by the machine learning techniques being used, as seen in the literature review. The present heart attack prediction algorithm also has a difficulty with the use of characteristics. Since the traits that should be used to predict heart attacks are common, the studies typically yield inaccurate results. In order to increase prediction accuracy, the suggested method aims to extract the appropriate attributes from datasets. Additionally, it will accurately diagnose the issue for customers so they can understand it without difficulty.

CHAPTER 7
MODULES DESCRIPTION

7.1 LIST OF MODULES

- MODULE 1: DATASETS ACQUISITION
- MODULE 2: PRE-PROCESSING
- MODULE 3: FEATURES SELECTION
- MODULE 4: CLASSIFICATION
- MODULE 5: DISEASE DIAGNOSIS

7.2 MODULE DESCRIPTION:

7.2.1 DATASETS ACQUISITION

A data set (or dataset, although this spelling is not present in many contemporary dictionaries like Merriam-Webster) is a collection of data. Most commonly a data set corresponds to the contents of a single database table, or a single statistical data matrix, where every column of the table represents a particular variable, and each row corresponds to a given member of the data set in question. The data set lists values for each of the variables, such as height and weight of an object, for each member of the data set. Each value is known as a datum. The data set may comprise data for one or more members, corresponding to the number of rows. The term data set may also be used more loosely, to refer to the data in a collection of closely related tables, corresponding to a particular experiment or event. In this module, we can upload the cardiovascular datasets related to heart diseases which includes the attributes such as age, gender, height, weight, systolic blood pressure, diastolic blood pressure, cholesterol, glucose, smoke, alcohol, active status, cardio labels.

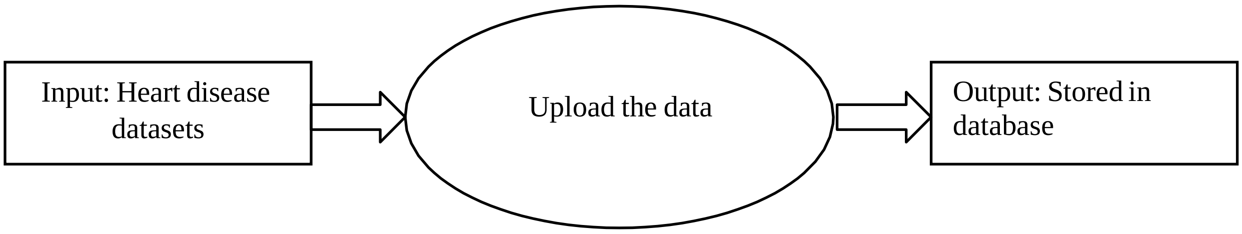


Fig 7.2.1 Dataset Acquisition diagram

7.2.2 PREPROCESSING

Data pre-processing is an important step in the [data mining] process. The phrase "garbage in, garbage out" is particularly applicable to data mining and machine learning projects. Data-gathering methods are often loosely controlled, resulting in out-of- range values, impossible data combinations, missing values, etc. Analysing data that has not been carefully screened for such problems can produce misleading results. Thus, the representation and quality of data is first and foremost before running an analysis. If there is much irrelevant and redundant information present or noisy and unreliable data, then knowledge discovery during the training phase is more difficult. Data preparation and filtering steps can take considerable amount of processing time. In this module, we can eliminate the irrelevant values and also estimate the missing values of data. Finally provide structured datasets.

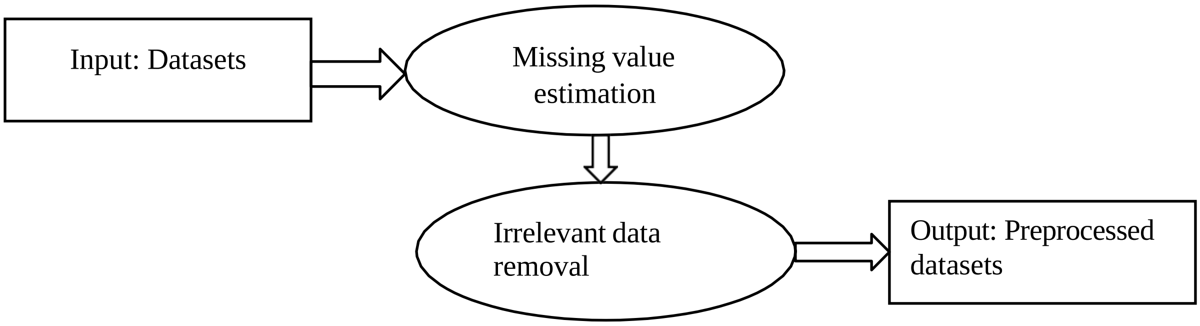


Fig 7.2.2 Preprocessing diagram

7.2.3 FEATURES SELECTION

Feature selection refers to the process of reducing the inputs for processing and analysis, or of finding the most meaningful inputs. A related term, feature engineering (or feature extraction), refers to the process of extracting useful information or features from existing data. Filter feature selection methods apply a statistical measure to assign a scoring to each feature. The features are ranked by the score and either selected to be kept or removed from the dataset. The methods are often uni-variate and consider the feature independently, or with regard to the dependent variable. It can be used to construct the multiple heart diseases. In this module, select the multiple features from uploaded datasets. And train the datasets with various disease labels such as Coronary heart diseases, Cardiac arrest, High blood pressure, Arrhythmia and normal.

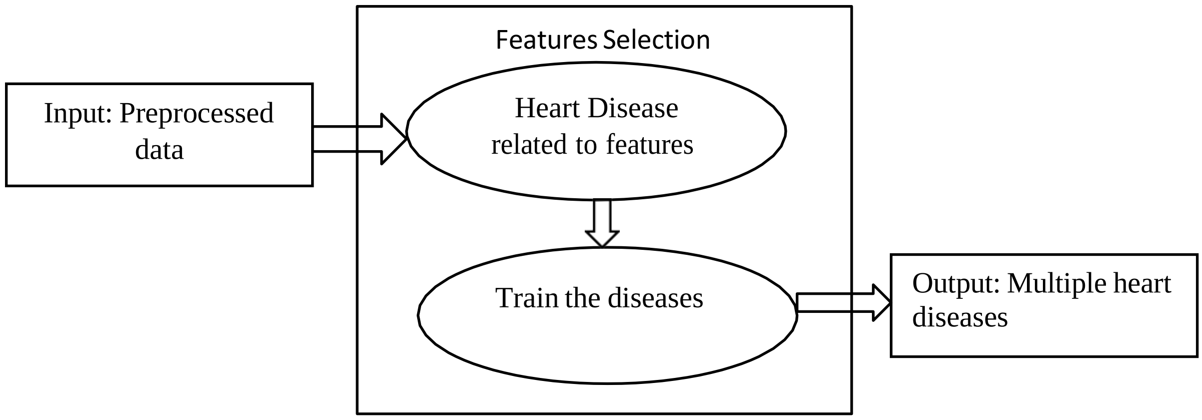


Fig 7.2.3 Features Selection diagram

7.2.4 CLASSIFICATION

In this module implement classification algorithm to predict the heart diseases. And using deep learning algorithm such as multi-layer perceptron algorithm to predict the diseases. A multilayer perceptron (MLP) is a feed forward artificial neural network model that maps sets of input data onto a set of appropriate outputs. It (MLP) consists of multiple layers of nodes in a directed graph, and each layer is fully connected to the next one. Each node is a neuron with a nonlinear activation function except for the input nodes. MLP utilizes a supervised learning technique called back propagation for training the network. MLP is a modified form of the standard linear perceptron and can distinguish data that are not linearly separable. If a multilayer perceptron (MLP) has a simple on-off mechanism i.e. linear activation function in all neurons, to determine whether or not a neuron fires, then it is easily proved with linear algebra that any number of layers can be reduced to the standard two-layer input-output model. The gradient techniques are then applied to the optimization methods to adjust the weights to minimize the loss function in the network. Hence, the algorithm requires a known and a desired output for all inputs in order to compute the gradient of loss function. Usually, the generalization of Multilayer Feed Forward Networks is done using delta rule which possibly makes a chain of iterative rules to compute gradients for each layer. Back Propagation Algorithm necessitates the activation function to be different between the neurons. The ongoing researches on parallel, distributed computing and computational neuroscience are currently implemented with the concepts of MultiLayer Perceptron using a Back Propagation Algorithm. The multilayer perceptron is the most known and most frequently used type of neural network. User can provide the features and automatically predict the diseases. A layered feed forward neural network is made up of layers, or subgroups of processing modules. Data is processed by a layer of processing components, which then transfers the results to a higher layer. The subsequent layer may then carry out its own calculations and transmit the outcomes to a subsequent layer. Last but not least, a subgroup of one or more processing elements determines the output of the network. Based on a weighted average of its inputs, each processing component performs its calculations. The first layer is the input layer, while the last layer is the output layer. The layers between these two are the ones that are concealed.

The steps in the neural network algorithm are as follows:

Step 1: Randomly initialise the weights and biases.

Step 2 is to feed the machine with the training sample.

Step 3: Continue using the inputs, and then compute each unit's net input and output in the hidden and output layers.

Reverse the defect to the concealed layer in step four.

Update the weights and biases to account for the spread of errors.

Step 5: The network's weights and biases are automatically adjusted using training and learning functions, which are mathematical operations.

Terminating condition in step six

Based on these phases, the neural network method outperforms traditional machine learning techniques.

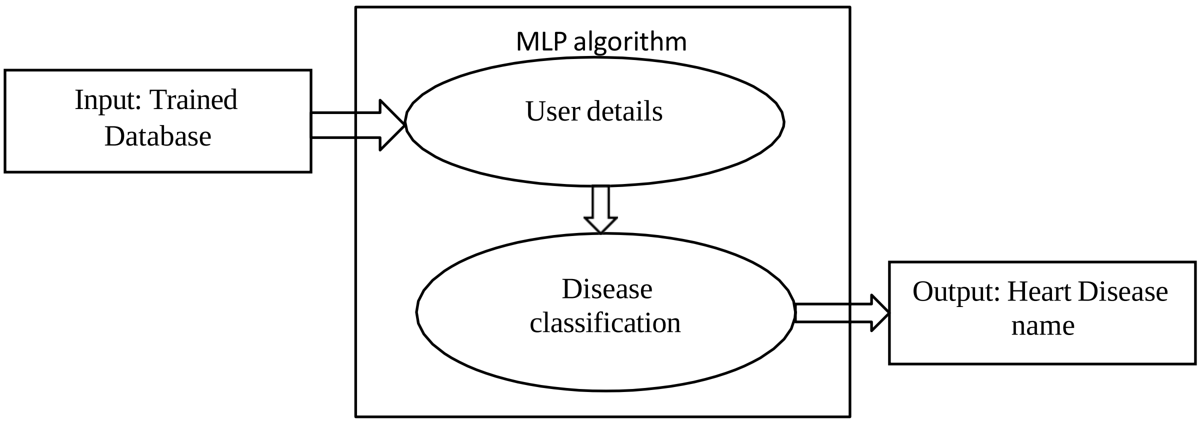


Fig 7.2.4 Classification diagram

7.2.5 DISEASE DIAGNOSIS

Medical decision support system is a decision-support program which is designed to assist physicians and other health professionals with decision making tasks, such as determining diagnosis of patients' data. In this module, provide the diagnosis information based on predicted heart diseases. Proposed system provides improved accuracy in heart disease prediction. Risk factors are conditions or habits that make a person more likely to develop a disease.

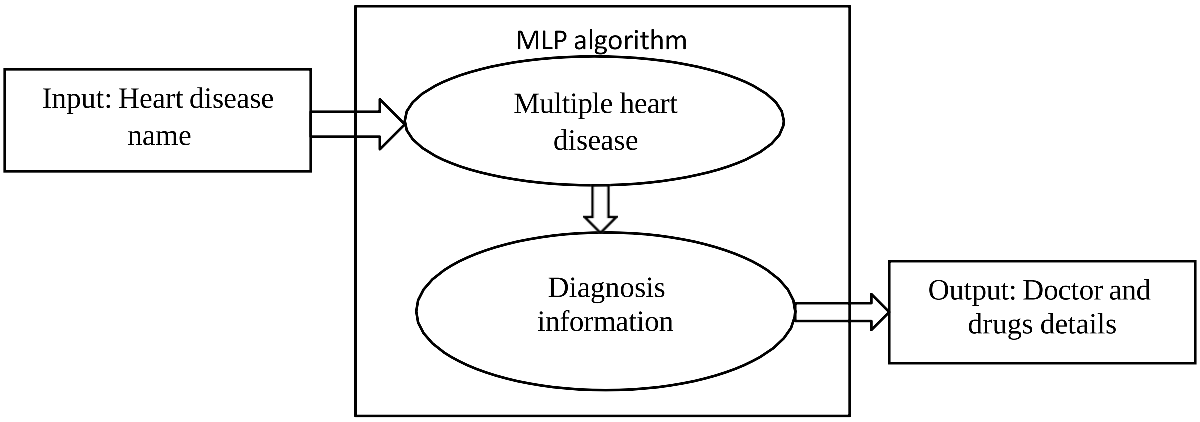


Fig 7.2.5 Disease Diagnosis diagram

CHAPTER 8

8. SYSTEM DESIGN

8.1 USECASE DIAGRAM

In this diagram, admin can upload the datasets that are collected from KAGGLE website. Then perform preprocessing steps to eliminate the irrelevant and missing datasets. Admin can train the features with x label and y label using multi-layer perceptron algorithm. User can input the datasets and predict the disease with precaution details and doctors

Fig 8.1 Use case diagram

8.2 CLASS DIAGRAM

A class diagram is a type of diagram used in software engineering and systems analysis to represent the structure of a system or software application. It is part of the Unified Modeling Language (UML), a standard modeling language used in software development. A class diagram is a type of diagram in the Unified Modeling Language (UML) that represents the structure of a system by showing its classes, their attributes, methods, and the relationships between them.

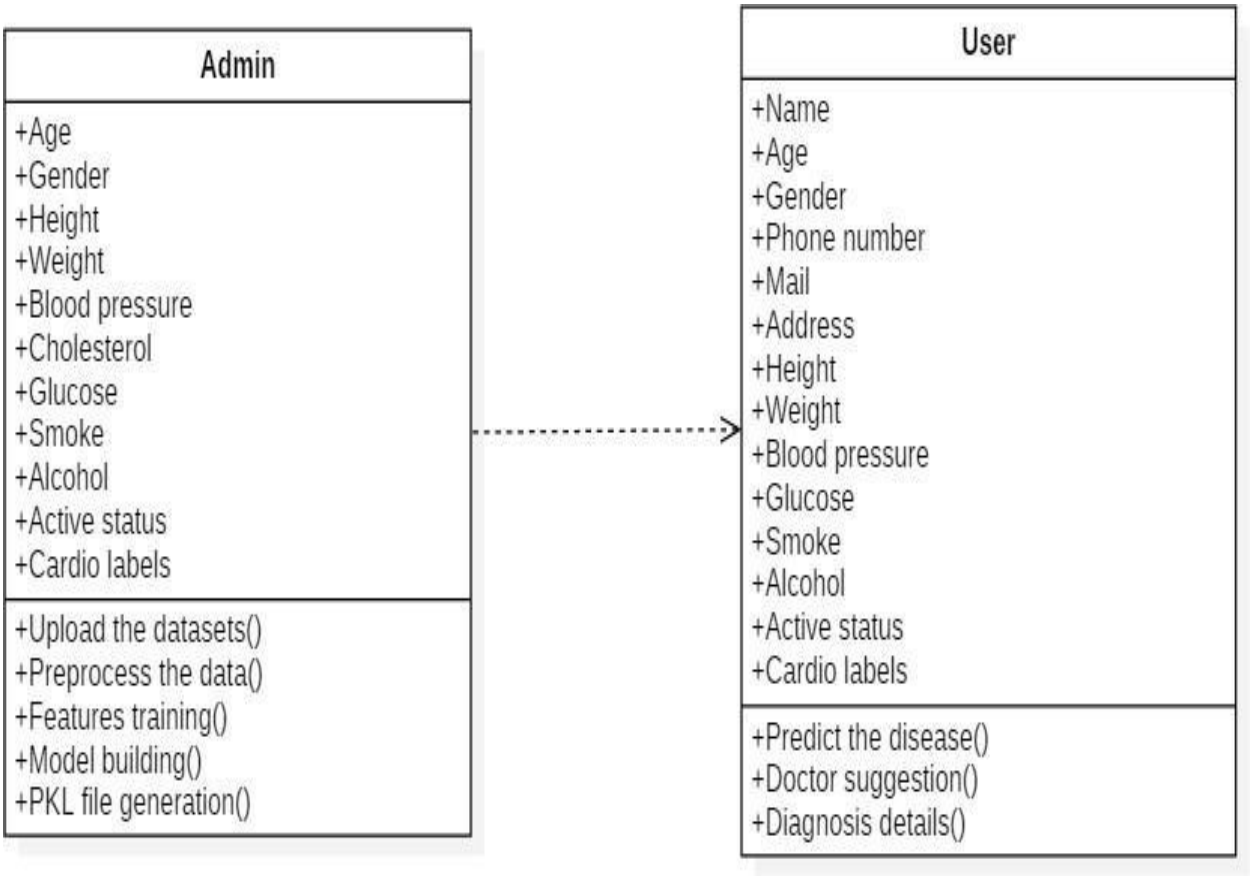


Fig 8.2 Class diagram

8.3 SEQUENCE DIAGRAM

A Sequence diagram is an interaction diagram that shows how objects operate with one another and in what order. It is a construct of a message sequence chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations in the Logical View of the system under development. Sequence diagrams are sometimes called event diagrams or event scenarios.

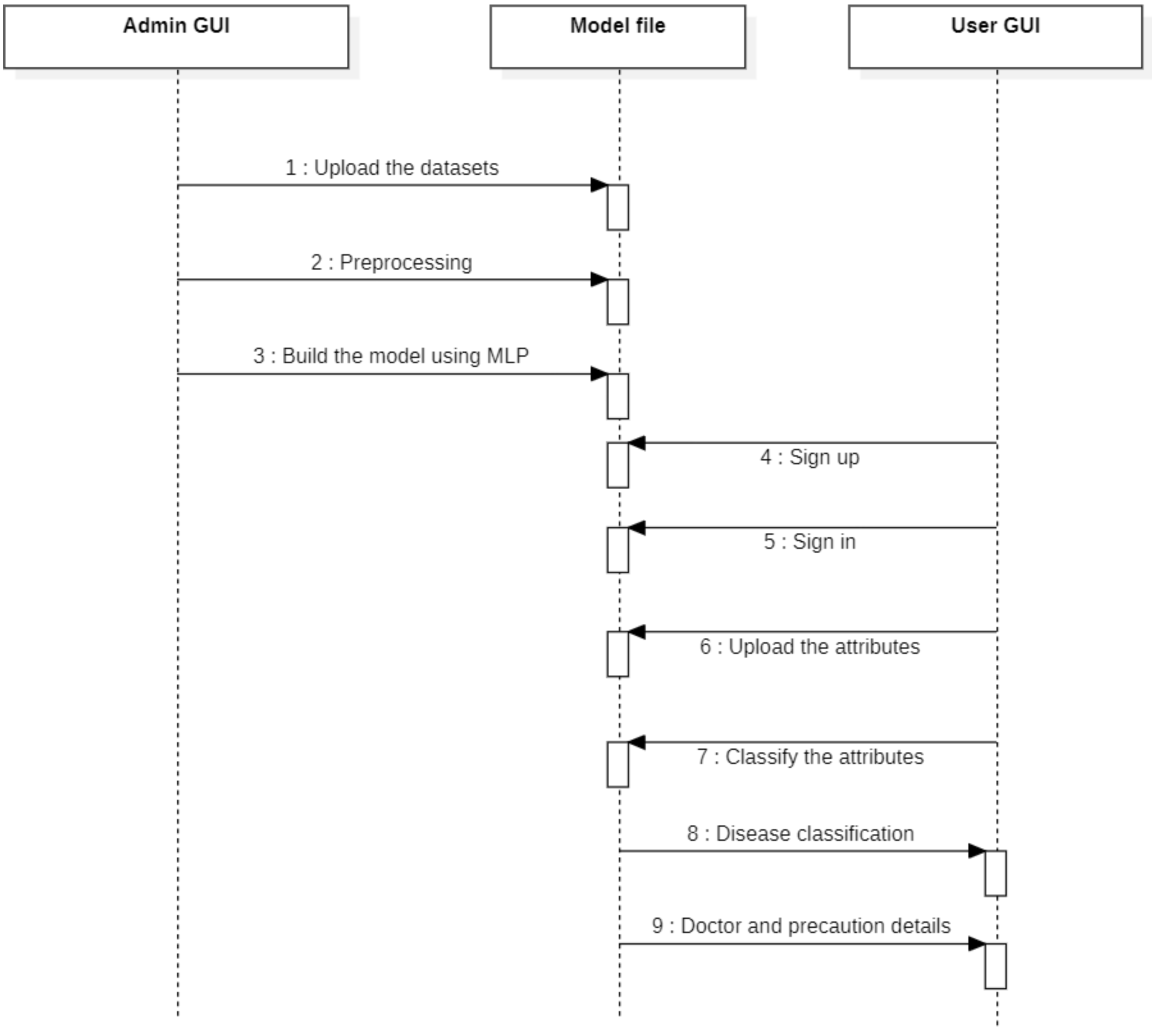


Fig 8.3 Sequence diagram

8.4 DEPLOYMENT DIAGRAM

A UML deployment diagram is a diagram that shows the configuration of run time processing nodes and the components that live on them. Deployment diagrams is a kind of structure diagram used in modelling the physical aspects of an object-oriented system. They are often be used to model the static deployment view of a system (topology of the hardware).

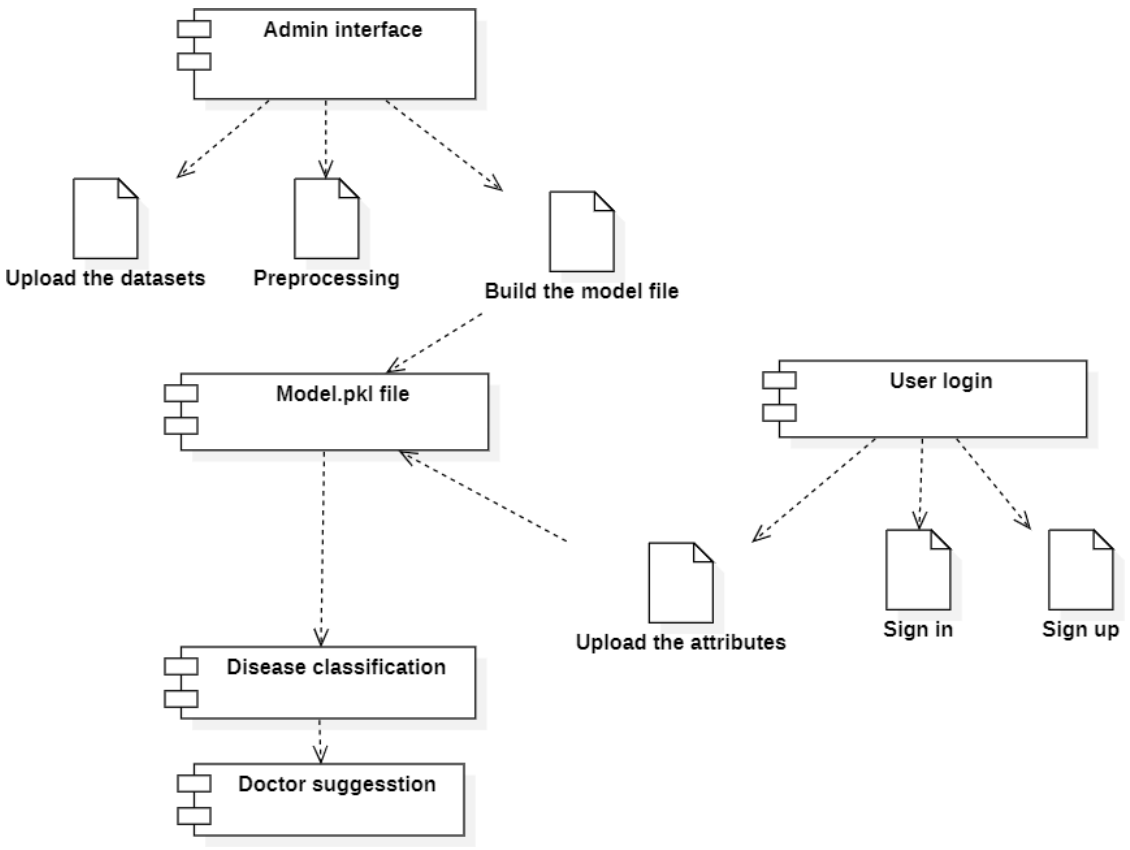


Fig 8.4 Deployment diagram

CHAPTER 9

9. SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

9.1 SOFTWARE TESTING STRATEGIES

Testing involves

- Unit Testing
- Functional Testing
- Acceptance Testing
- Validation Testing

9.1.1 UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

9.1.2 FUNCTIONAL TESTING

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must beexercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

9.1.3 ACCEPTANCE TESTING

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

9.1.4 VALIDATION TESTING

At the culmination of integration testing, software is completely assembled as a package, interfacing errors have been uncovered and corrected, and a final series of software tests-validation testing begin. Validation can be defined in many ways, but a simple definition is that validation succeeds when software functions in a manner that can be reasonably expected by the customer.

CHAPTER 10

10. CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

In this project the problem of constraining and summarizing different algorithms of data mining used in the field of medical prediction are discussed. The focus is on using different algorithms and combinations of several target attributes for intelligent and effective heart disease prediction using data mining. Data mining technology provides an important means for extracting valuable medical rules hidden in medical data and acts as an important role in disease prediction and clinical diagnosis. There is an increasing interest in using classification to identify disease which is present or not. In the current study, have demonstrated, using a large sample of patients hospitalized with classification. Classification algorithm is very sensitive to noisy data. If any noisy data is present then it causes very serious problems regarding to the processing power of classification. It not only slows down the task of classification algorithm but also degrades its performance. Hence, before applying classification algorithm it must be necessary to remove all those attributes from datasets who later on acts as noisy attributes. In this research work, we can implement pre-processing steps and implemented the classification rule algorithms namely multi-layer perceptron is used for classifying datasets which are uploaded by user. By analysing the experimental results, it is observed that the multi-layer perceptron technique has yields better result than other techniques.

10.2 FUTURE ENHANCEMENT

In future we tend to improve efficiency of performance by applying other data mining techniques and algorithms.

APPENDICES

APPENDIX A: SAMPLE CODE

APP.PY

```
from flask import Flask, render_template, flash, request, session, send_file
import pickle
import numpy as np
import mysql.connector
import sys

app = Flask(_name_)
app.config['DEBUG']
app.config['SECRET_KEY'] = '7d441f27d441f27567d441f2b6176a'



@app.route("/")
def homepage():
    return render_template('index.html')



@app.route("/Home")
def Home():
    return render_template('index.html')



@app.route("/AdminLogin")
def AdminLogin():
    return render_template('AdminLogin.html')



@app.route("/NewDoctor")
```

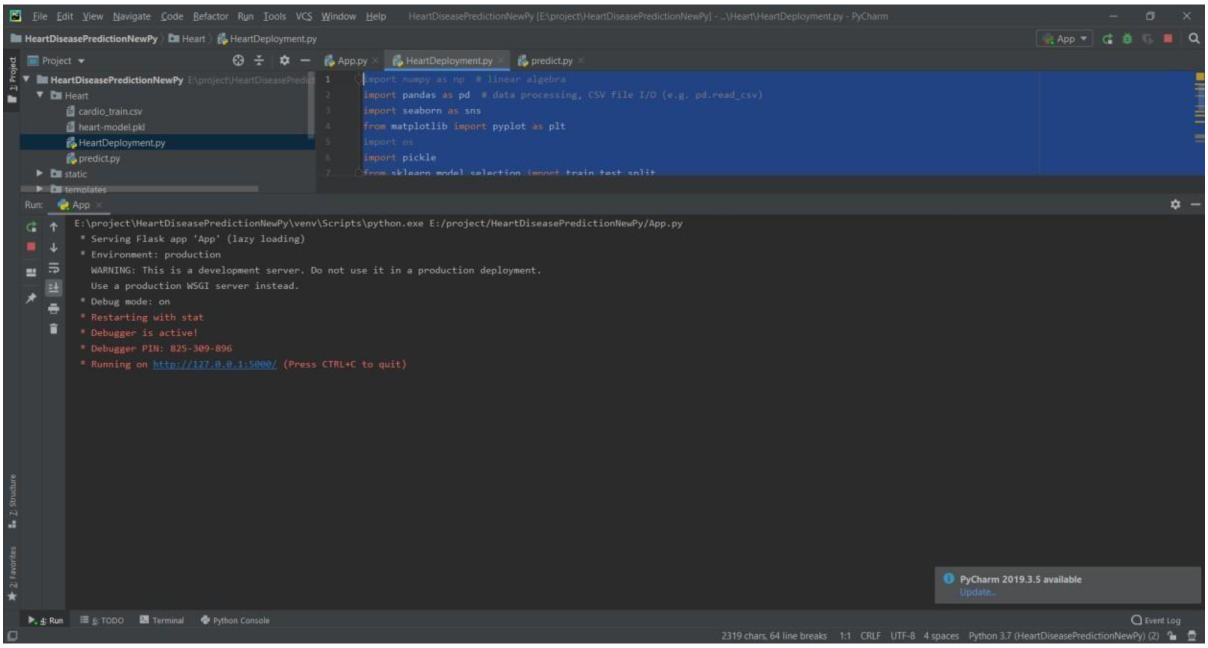
```
def NewDoctor():
    return render_template('NewDoctor.html')



@app.route("/DoctorLogin")
def DoctorLogin():
    return render_template('DoctorLogin.html')



@app.route("/UserLogin")
def UserLogin():
    return render_template('UserLogin.html')



@app.route("/NewUser")
def NewUser():
    return render_template('NewUser.html')



@app.route("/AdminHome")
def AdminHome():
    conn = mysql.connector.connect(user='root', password='', host='localhost',
database='5heartbd')
    cur = conn.cursor()
    cur.execute("SELECT * FROM regtb  ")
    data = cur.fetchall()
    return render_template('AdminHome.html', data=data)



@app.route("/adminlogin", methods=['GET', 'POST'])
def adminlogin():
```

```
    if request.method == 'POST':
        if request.form['uname'] == 'admin' and request.form['password'] == 'admin':

            conn = mysql.connector.connect(user='root', password='', host='localhost',
database='5heartbd')
            cur = conn.cursor()
            cur.execute("SELECT * FROM regtb ")
            data = cur.fetchall()
            flash("Login successfully")
            return render_template('AdminHome.html', data=data)

        else:
            flash("UserName Or Password Incorrect!")
            return render_template('AdminLogin.html')


@app.route("/DoctorInfo")
def DoctorInfo():
    conn = mysql.connector.connect(user='root', password='', host='localhost',
database='5heartbd')
    cur = conn.cursor()
    cur.execute("SELECT * FROM doctortb  ")
    data = cur.fetchall()
    return render_template('DoctorInfo.html', data=data)


@app.route("/ADRemove")
def ADRemove():
    id = request.args.get('id')
    conn = mysql.connector.connect(user='root', password='', host='localhost',
database='5heartbd')
```

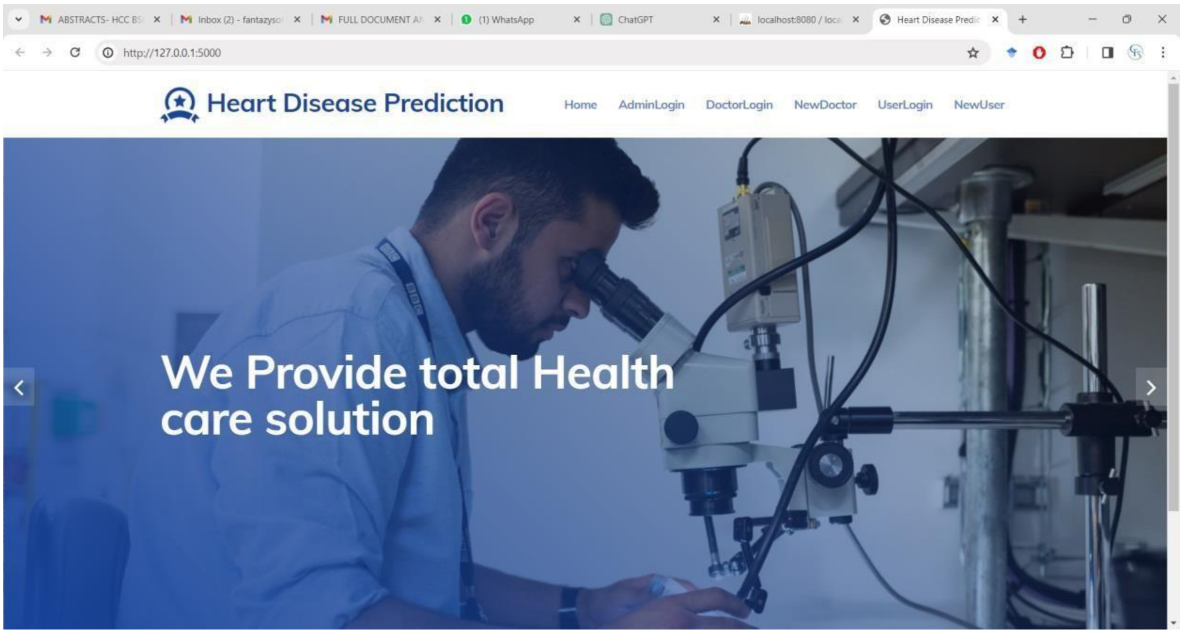
```
    cursor = conn.cursor()
    cursor.execute(
        "delete from doctortb where id='" + id + "'")
    conn.commit()
    conn.close()

    flash('Doctor  info Remove Successfully!')

    conn = mysql.connector.connect(user='root', password='', host='localhost',
database='5heartbd')
    cur = conn.cursor()
    cur.execute("SELECT * FROM doctortb  ")
    data = cur.fetchall()
    return render_template('DoctorInfo.html', data=data)


@app.route("/AURemove")
def AURemove():
    id = request.args.get('id')
    conn = mysql.connector.connect(user='root', password='', host='localhost',
database='5heartbd')
    cursor = conn.cursor()
    cursor.execute(
        "delete from regtb where id='" + id + "'")
    conn.commit()
    conn.close()
```

APPENDIX B: SCREENSHOTS

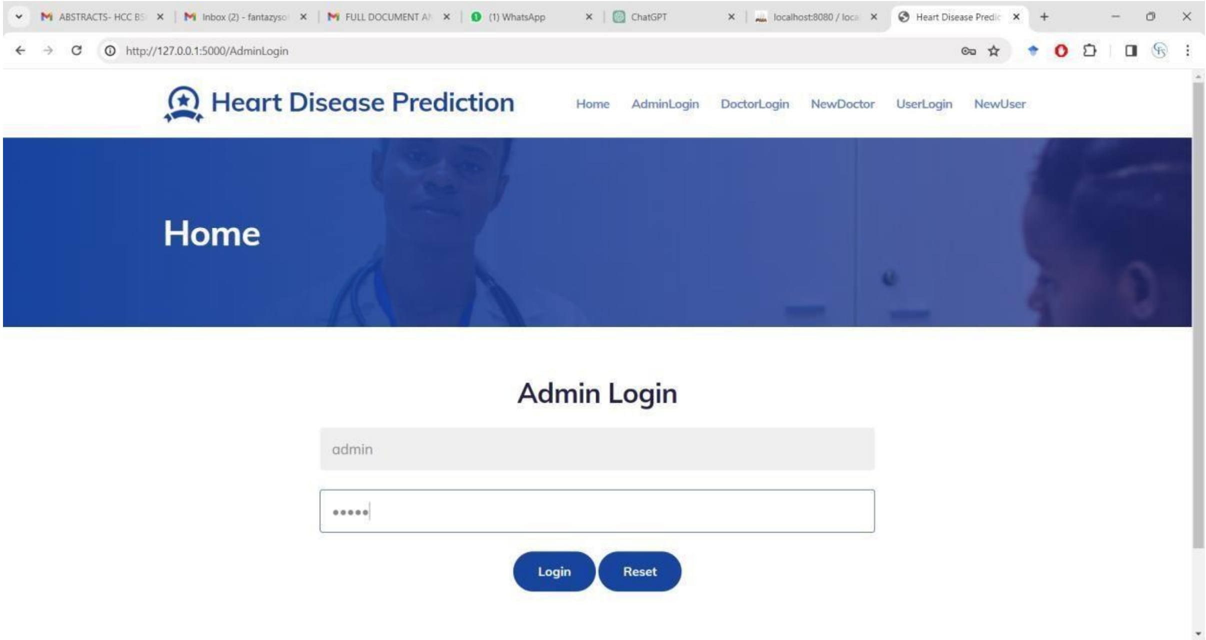
EXECUTION PAGE

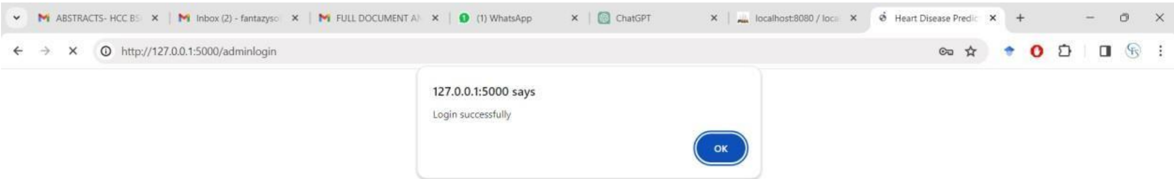


HOME PAGE



ADMIN LOGIN


http://127.0.0.1:5000/AdminLogin
Heart Disease Prediction
Home
AdminLogin
DoctorLogin
NewDoctor
UserLogin
NewUser
Home
Admin Login
admin
Login
Reset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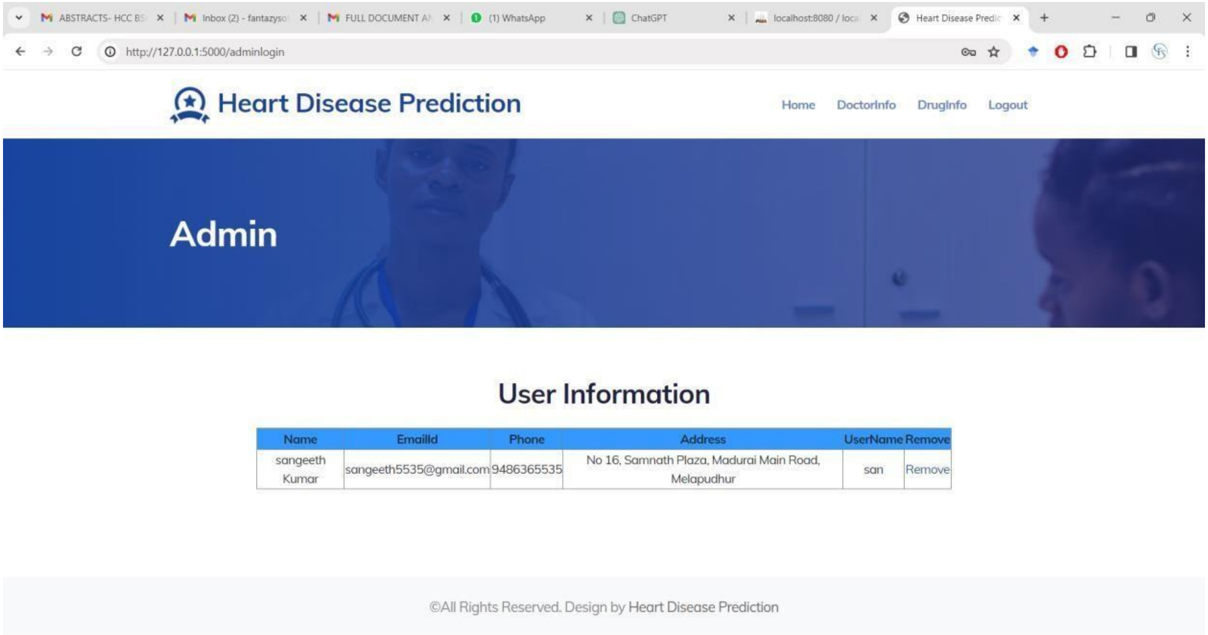
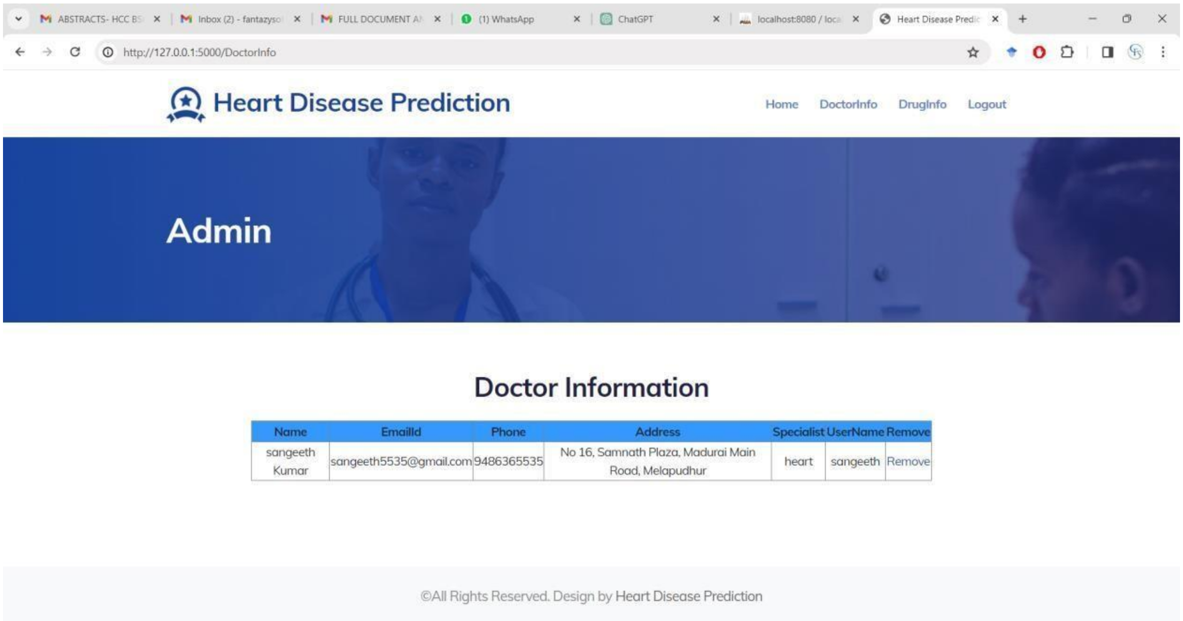
Login successfully
OK

USER INFORMATION



DOCTOR INFORMATION



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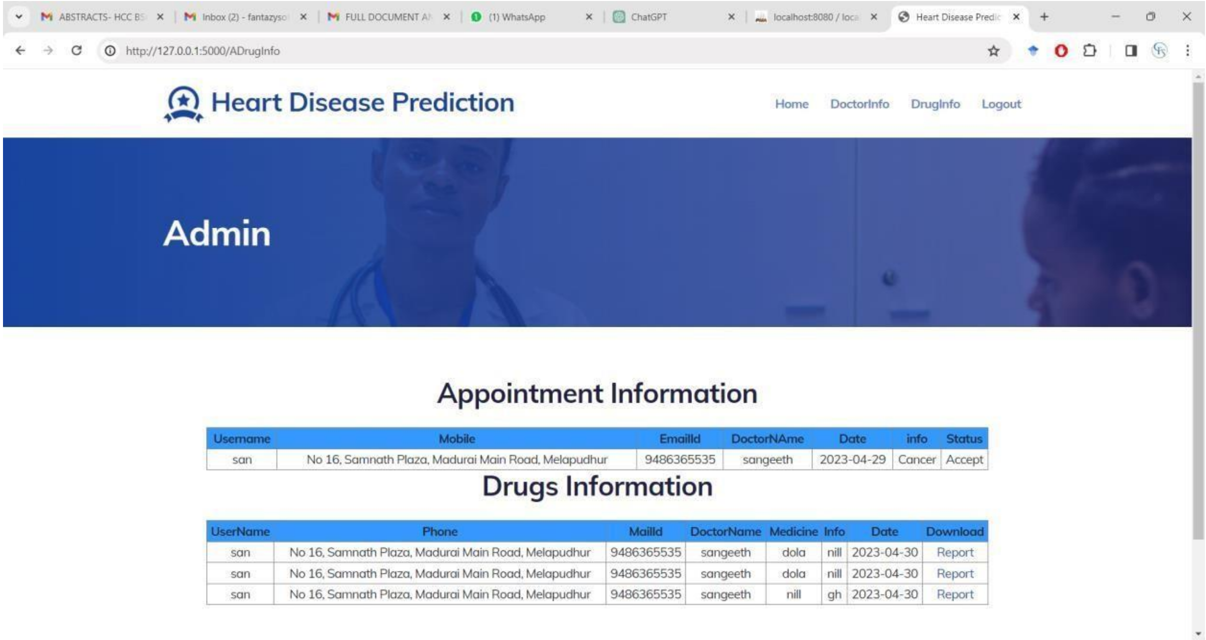
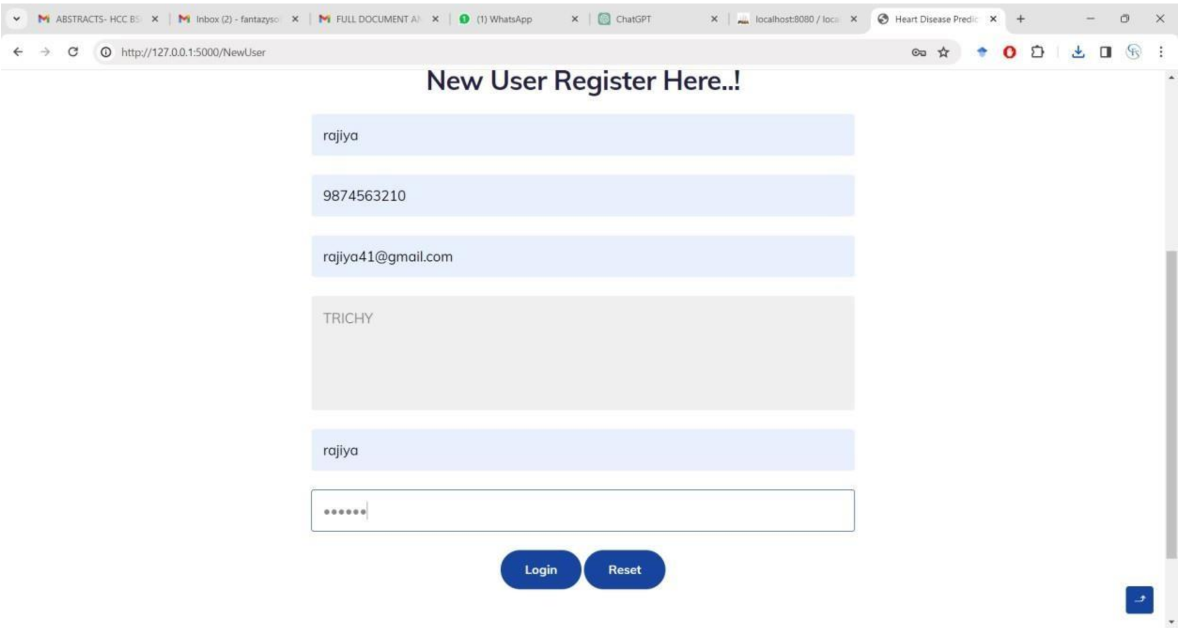
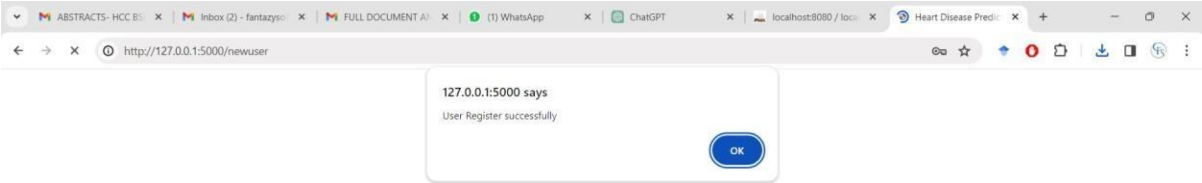


NEW USER REGISTER




ABSTRACTS- HCC BS
Inbox (2) - fantazyso
FULL DOCUMENT A
(1) WhatsApp
ChatGPT
localhost:8080 / loca
Heart Disease Predic
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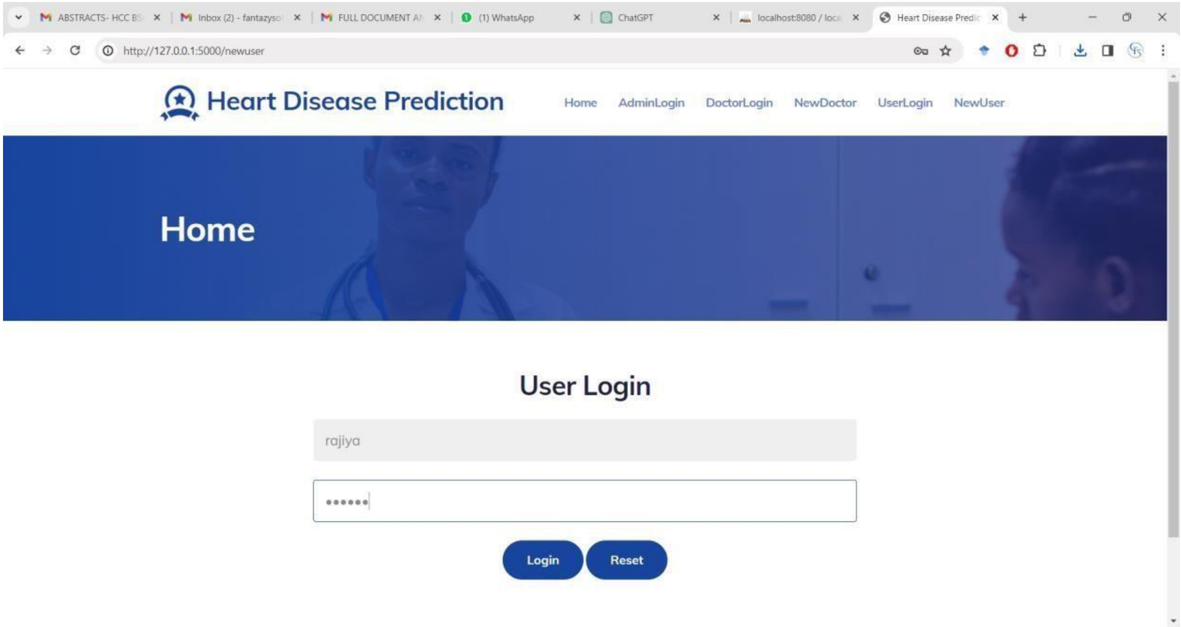
127.0.0.1:5000 says
User Register successfully
OK

USER LOGIN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http://127.0.0.1:5000/newuser
Heart Disease Prediction
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AdminLogin
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NewDoctor
UserLogin
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Home
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Login
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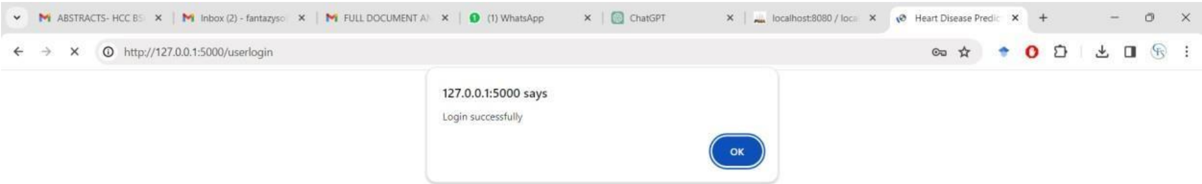

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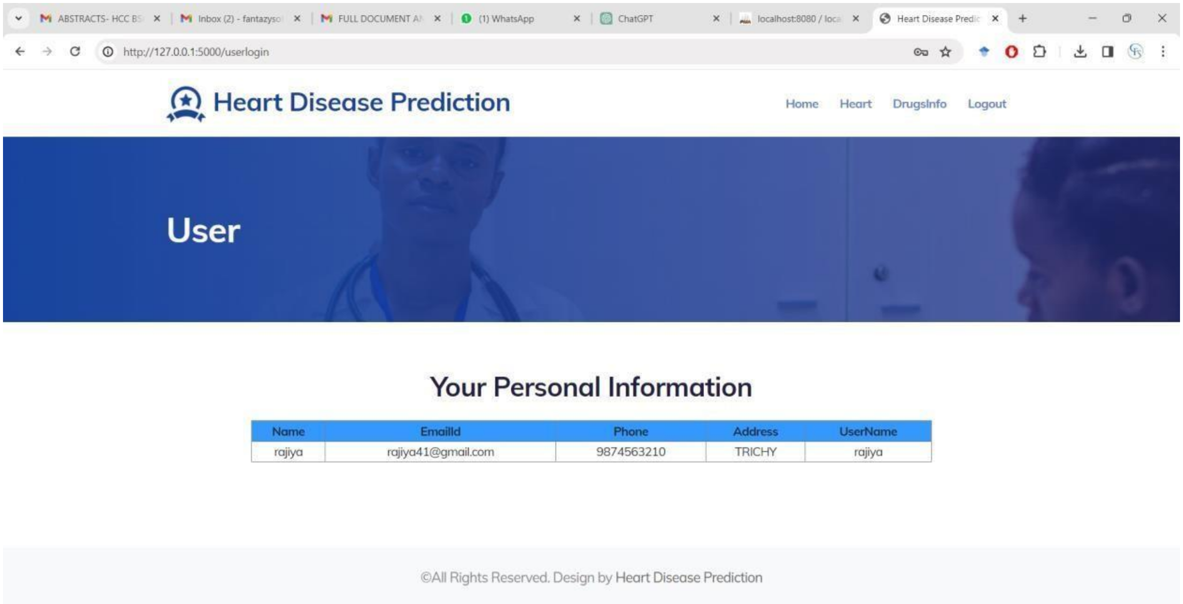
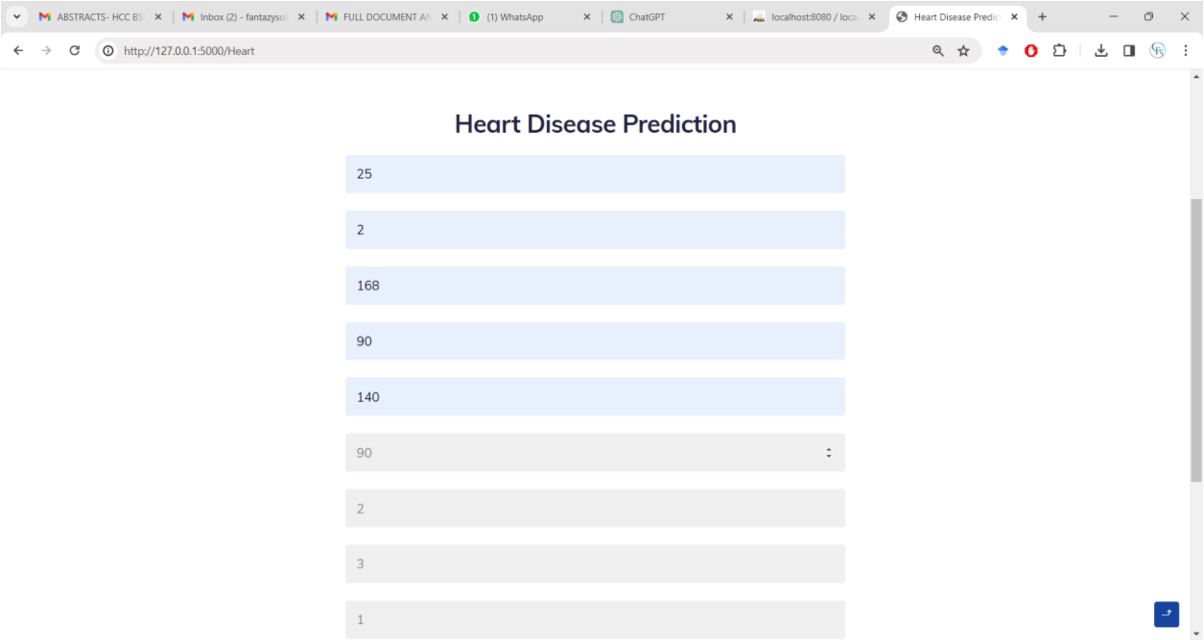
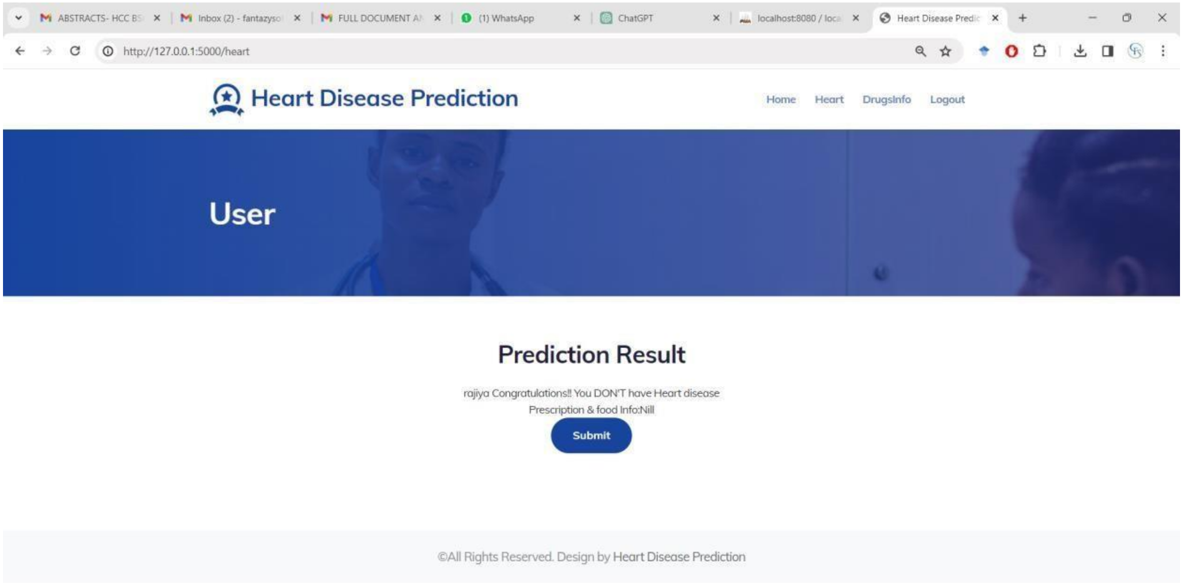
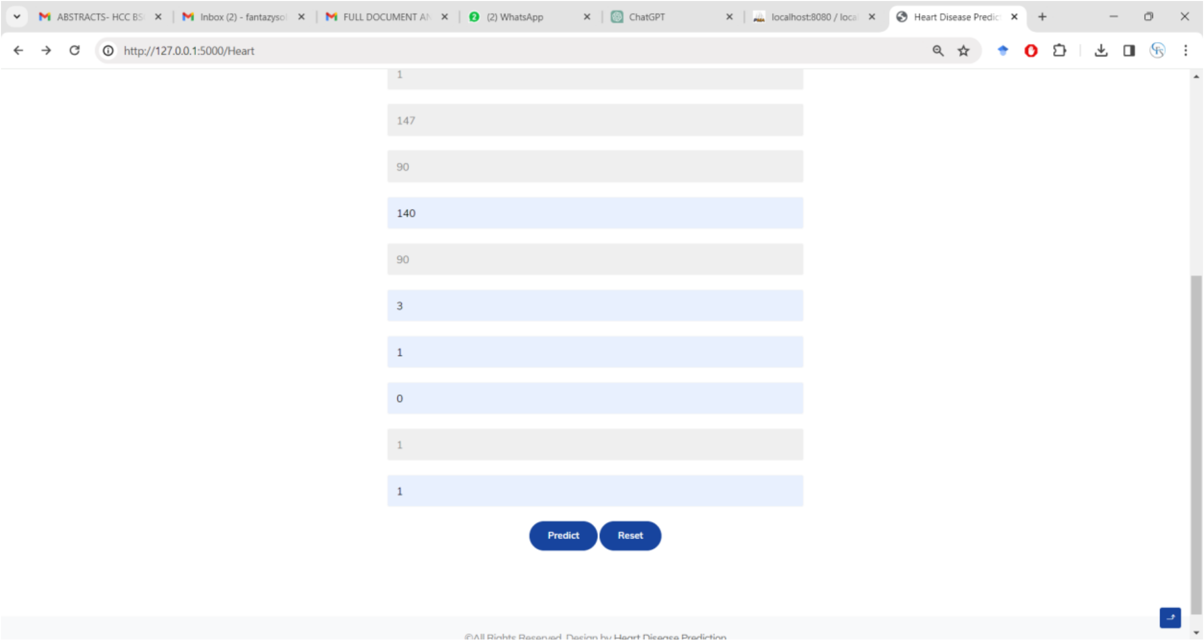


HEART DISEASE PREDICTION



PREDICTION RESULT

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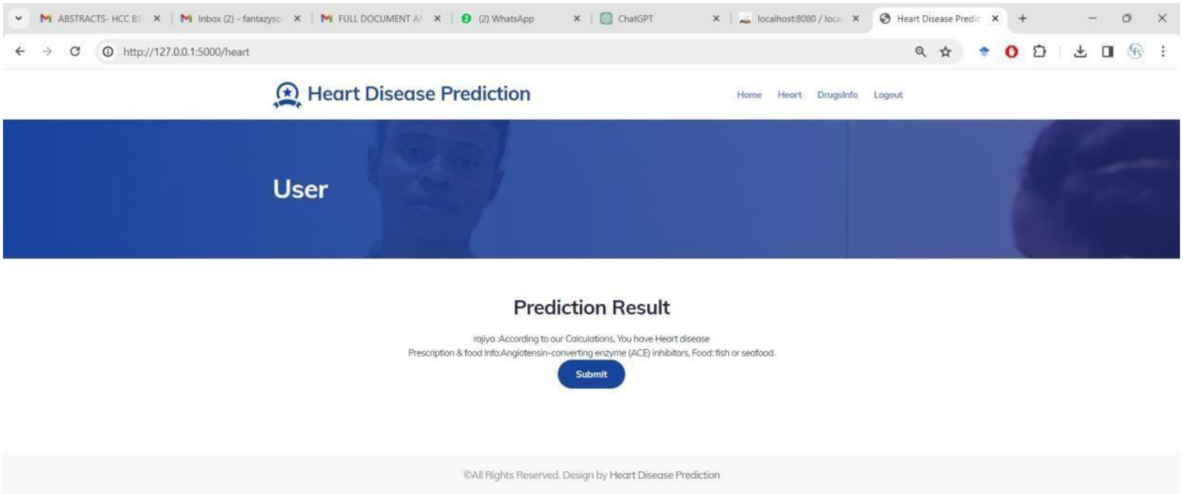
http://127.0.0.1:5000/heart
Heart Disease Prediction
Home
Heart
DrugsInfo
Logout
User
Prediction Result
rajiya :According to our Calculations, You have Heart disease
Prescription & food Info:Angiotensin-converting enzyme (ACE) inhibitors, Food: fish or seafood.
Submit
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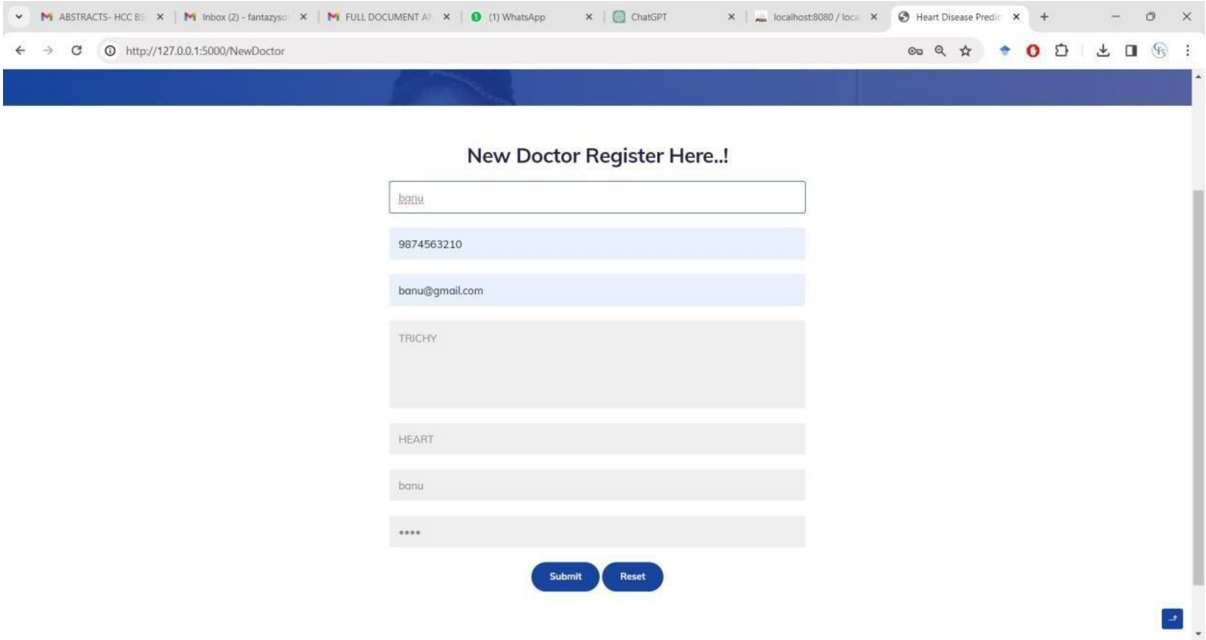
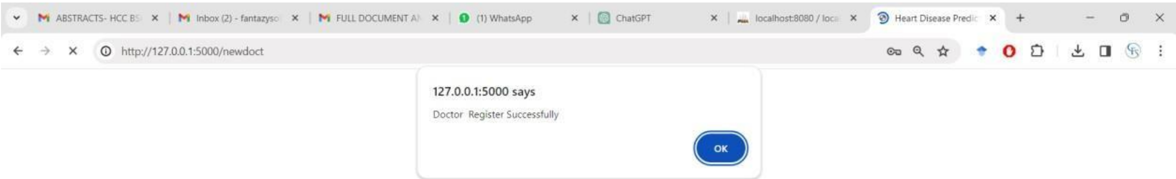
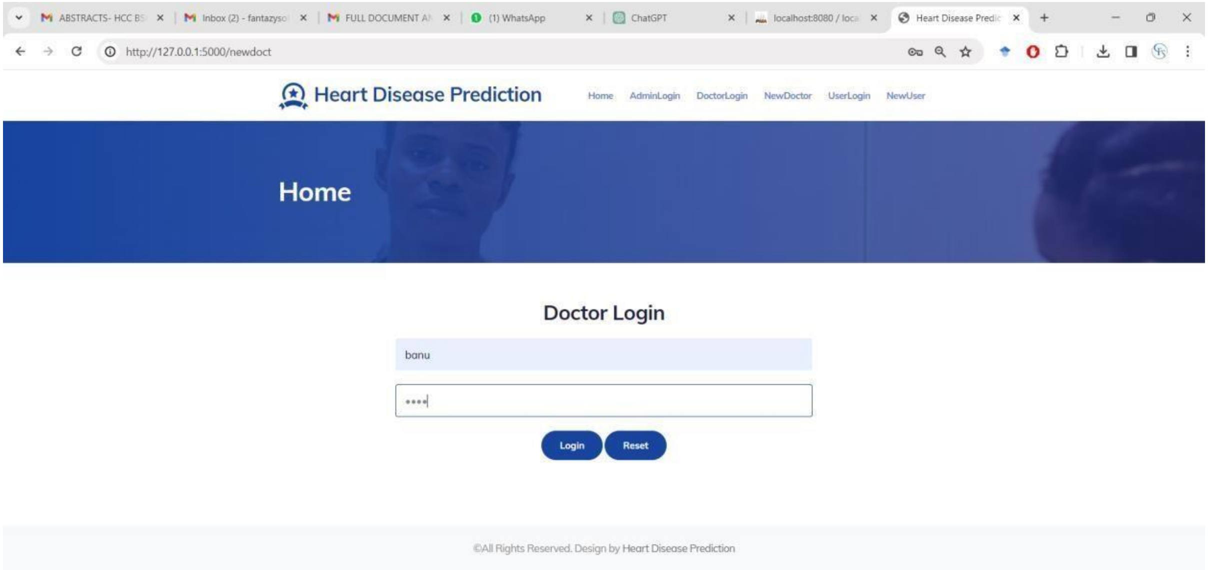
DOCTOR REGISTER



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AdminLogin
DoctorLogin
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UserLogin
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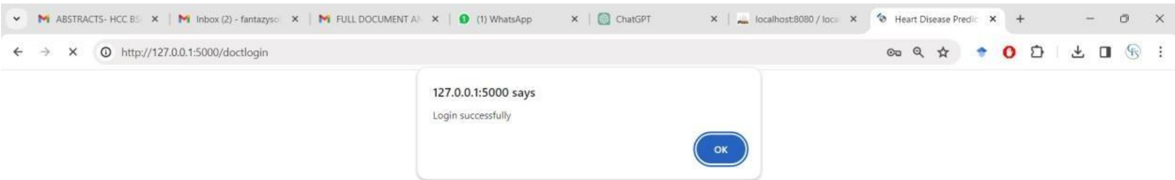
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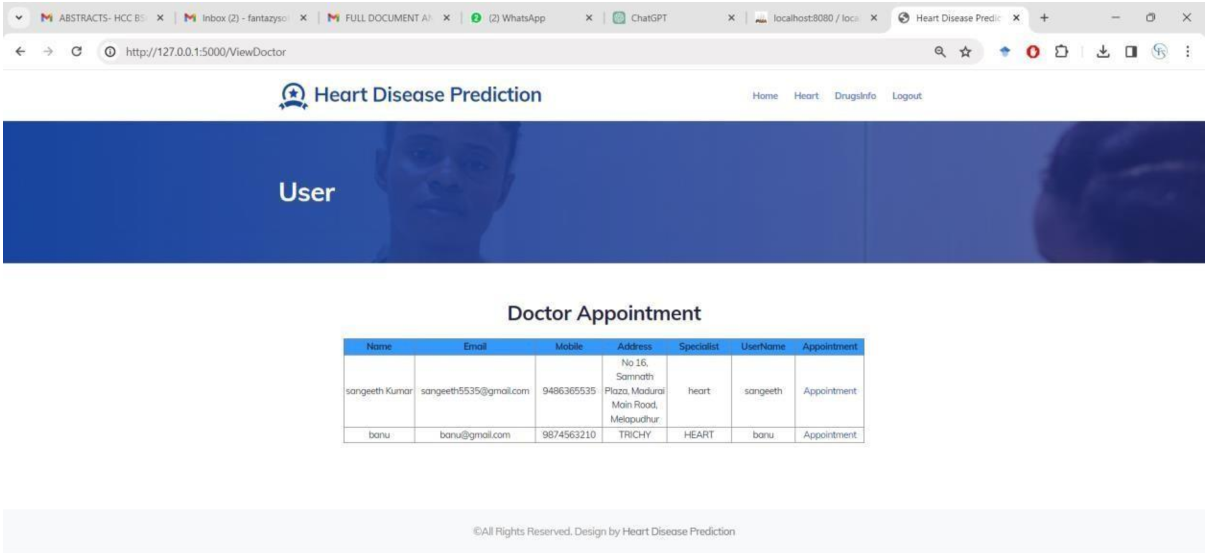
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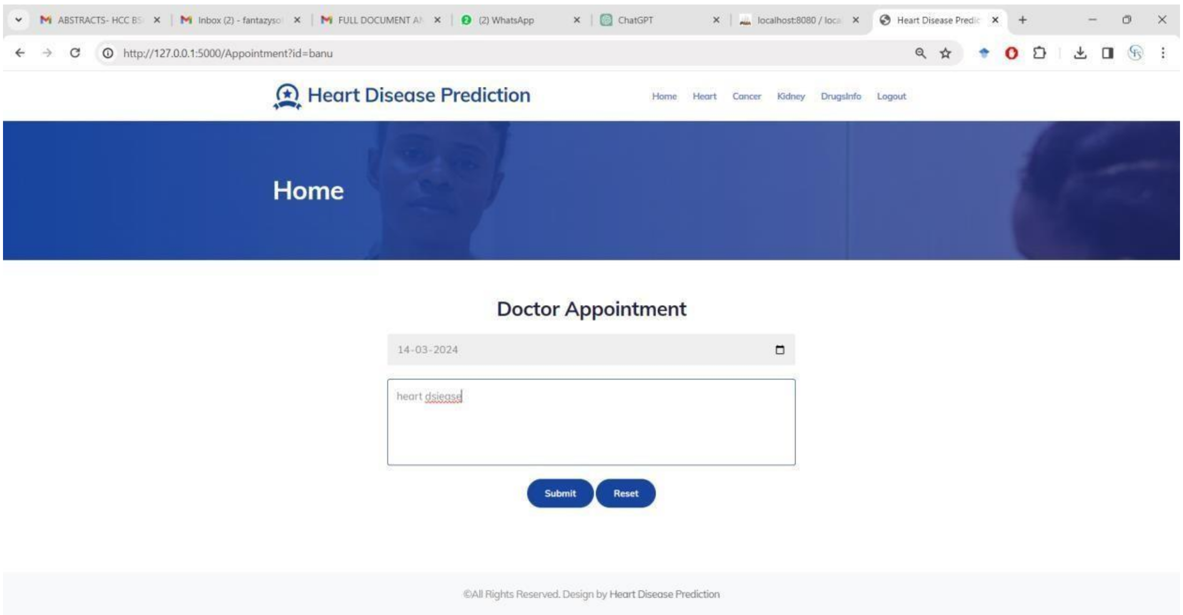
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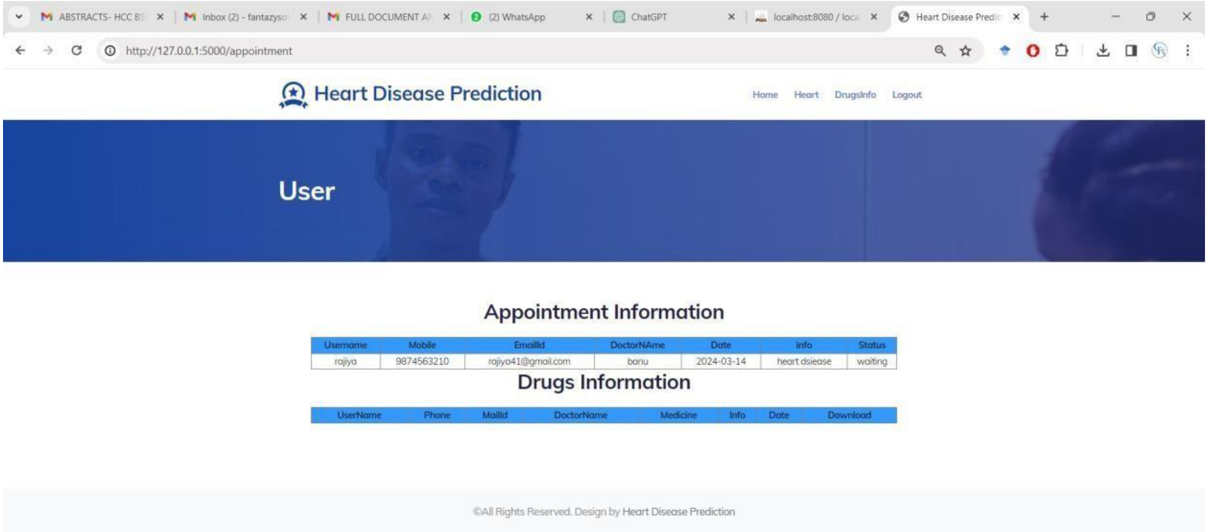
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Appointment Information
Username Mobile EmailId DoctorNAme Date Info Status
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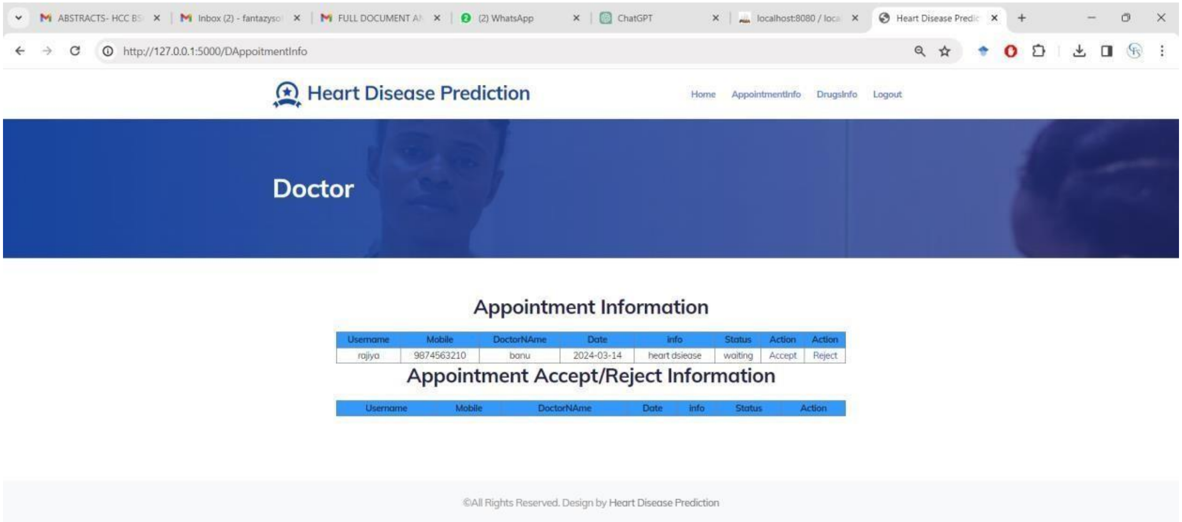
Drugs Information
UserName Phone MailId DoctorName Medicine Info Date Download
©All Rights Reserved. Design by Heart Disease Prediction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Username Mobile DoctorNAme Date Info Status Action
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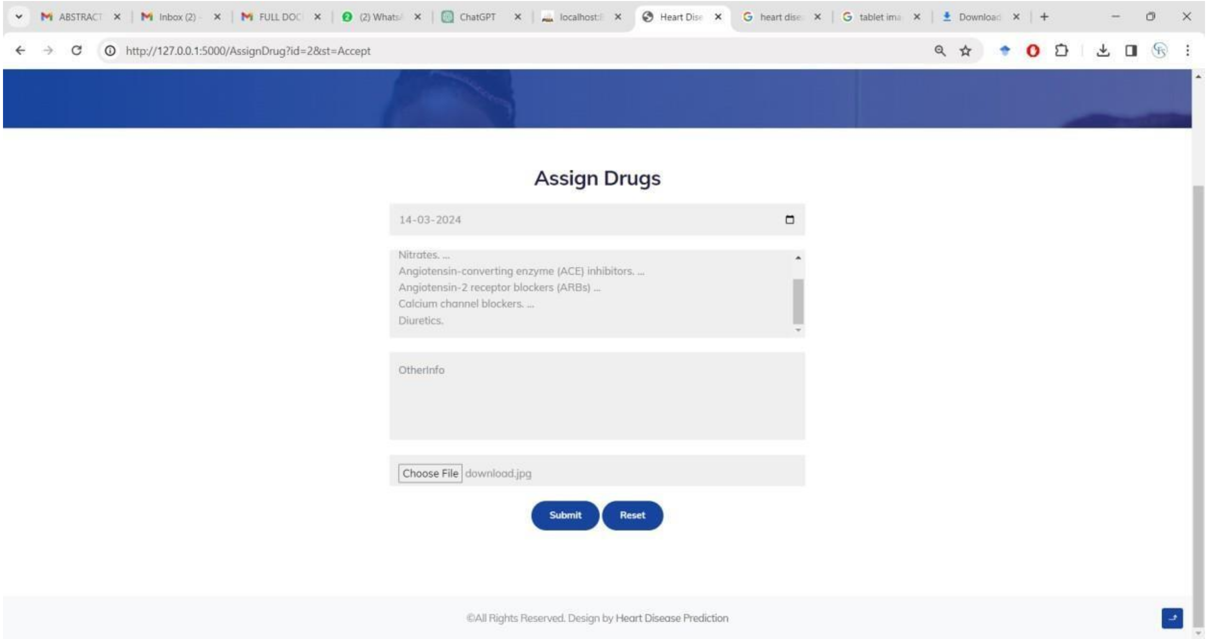
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Nitrates. ...
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Angiotensin-2 receptor blockers (ARBs) ...
Calcium channel blockers. ...
Diuretics.
OtherInfo
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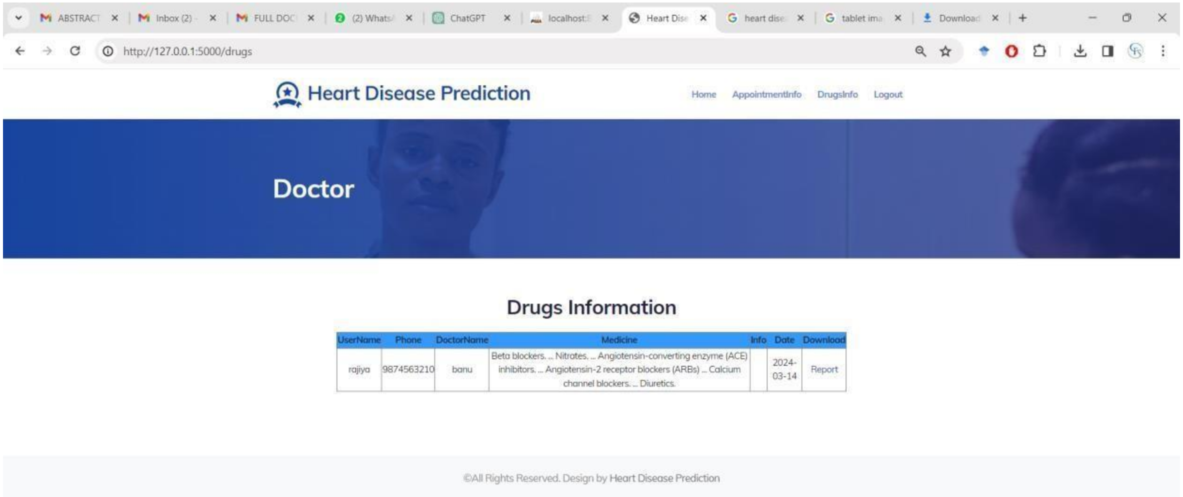
Reset
©All Rights Reserved. Design by Heart Disease Prediction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Heart Disease Prediction
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rajiya 9874563210 banu Beta blockers. ... Nitrates. ... Angiotensin-converting enzyme (ACE) inhibitors. ... Angiotensin-2 receptor blockers (ARBs) ... Calcium channel blockers. ... Diuretics. 2024-03-14 Report
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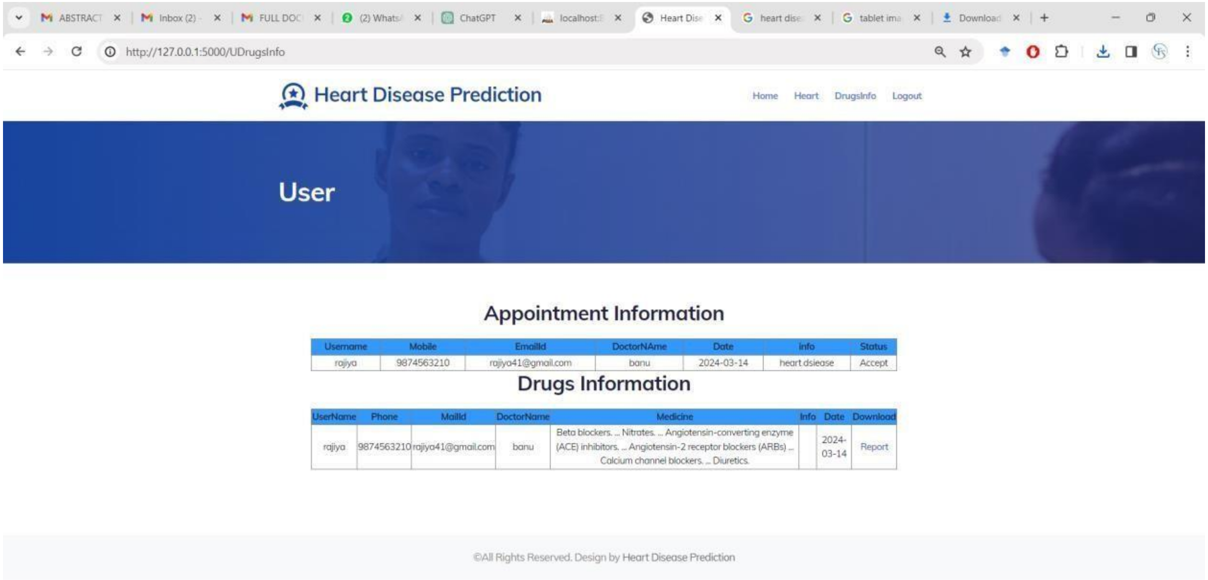
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Drugs Information
UserName Phone MailId DoctorName Medicine Info Date Download
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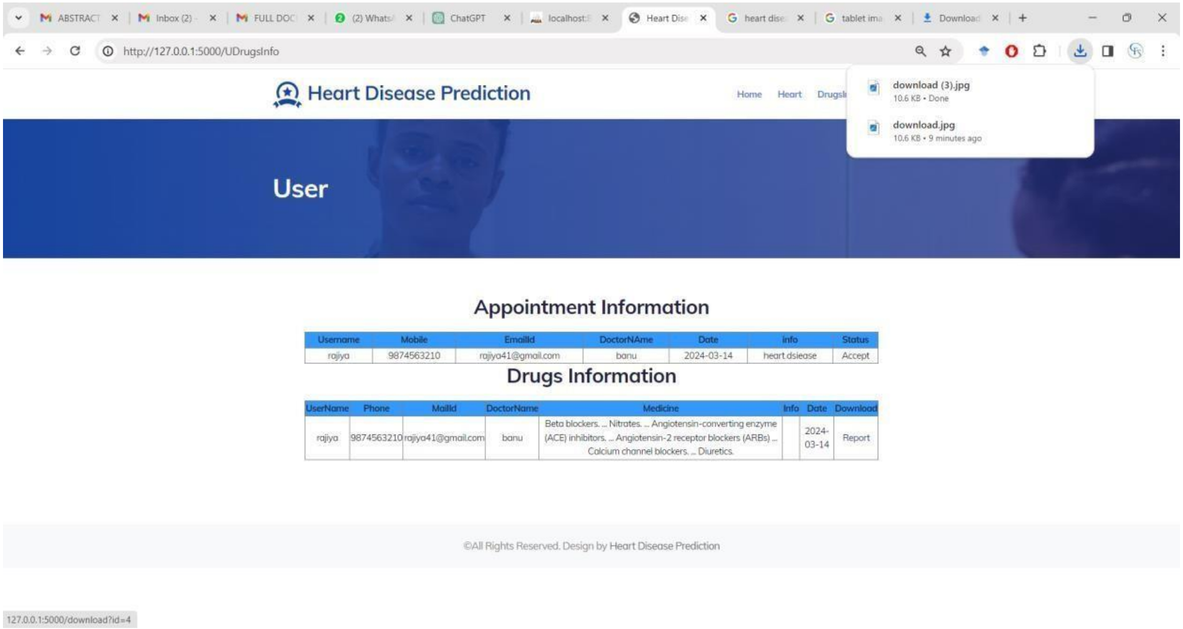
http://127.0.0.1:5000/UDrugsInfo
Heart Disease Prediction
Home
Heart
download (3).jpg
10.6 KB • Done
download.jpg
10.6 KB • 9 minutes ago
User
Appointment Information
Username Mobile EmailId DoctorNAme Date info Status
rajiya 9874563210 rajiya41@gmail.com banu 2024-03-14 heart dsiease Accept
Drugs Information
UserName Phone MailId DoctorName Medicine Info Date Download
rajiya 9874563210 rajiya41@gmail.com banu Beta blockers. ... Nitrates. ... Angiotensin-converting enzyme (ACE) inhibitors. ... Angiotensin-2 receptor blockers (ARBs) ... Calcium channel blockers. ... Diuretics. 2024-03-14 Report
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CHAPTER 11
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